

Overreaction to Scarcity: Precautionary Demand and Oil Price Reversals

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Abstract

Text-based measures of attention to oil demand and supply in the financial press *both* predict crude oil futures returns negatively over the following year while leading analysts to revise their price forecasts upward, the signature of belief overreaction. In contrast, the real economic activity index of [Kilian \(2009\)](#), the canonical measure of realized demand, positively predicts future returns and analyst forecast revisions. To locate the difference we decompose the analytical content of the EIA’s monthly *Short-Term Energy Outlook* into the structural channels of [Kilian \(2009\)](#) using a two-stage large language model pipeline, reading directly the precautionary demand that the structural literature identifies as an unobservable expectations shock. Aggregate demand and supply content are updated rationally: they move forecasts, leave subsequent returns unaffected, and move production, inventories, and drilling as a structural framework would predict. Precautionary demand content moves forecasts up and subsequent returns down, forecasts returns out of sample, and is followed within a quarter by net selling from speculators and reduced hedging by producers. Expectations about future scarcity, rather than realized demand or supply, drive the oil price movements that subsequently reverse.

Keywords: Oil prices; precautionary demand; diagnostic expectations; overreaction; large language models; speculative positioning; commodity markets.

JEL Codes: G12; G14; G41; Q41.

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1 Introduction

A persistent gap separates the structural analysis of oil prices from the beliefs that move them. Where [Hamilton \(1983, 2003\)](#) attributes oil price shocks to exogenous supply disruptions, [Kilian \(2009\)](#) shows that most fluctuations are driven by demand, and within demand by a precautionary component driven by expectations of future scarcity. Despite the framework’s central role ([Kilian and Murphy, 2014](#); [Baumeister and Hamilton, 2019](#)), evidence of the effect of supply versus demand on oil prices remains mixed ([Hamilton, 2009](#)), and the precautionary component of demand is identified only as a residual, because expectations are not observed. A separate literature documents that beliefs overreact to news across asset markets ([Bordalo, Gennaioli, and Shleifer, 2018](#); [Bordalo, Gennaioli, La Porta, and Shleifer, 2024](#)), and expectations of future scarcity are the kind of belief that framework predicts will overshoot. This paper argues that the price movements attributed to precautionary demand generate overreactions. News about expected scarcity, not news about realized supply or demand, predicts subsequent reversals in oil prices.

Using keyword dictionaries derived from the Energy Information Administration’s own analytical language, we measure the share of newspaper coverage of oil devoted to rising demand and to rising supply, from 1986 through 2025. Both measures predict crude oil futures returns negatively: a one-standard-deviation increase in demand attention is followed by a cumulative return of -3.1% over six months and -5.7% over twelve ($t = -2.77$ and -3.44), and supply attention by -3.8% and -4.5% ($t = -3.33$ and -2.71). At the same time both predict upward revisions in the consensus analyst forecast of the oil price, by 5.6% and 4.1% per standard deviation ($t = 5.33$ and 2.62). Forecasts move in the direction the news warrants but prices subsequently move against them, the signature of overreaction in the sense of [Bordalo, Gennaioli, and Shleifer \(2018\)](#). The canonical measure of realized demand does the opposite. The real economic activity index of [Kilian \(2009\)](#) predicts upward forecast revisions of the same size (4.2% , $t = 6.18$) and *higher* subsequent returns, rising to 4.3% over twelve months ($t = 2.55$), which is the pattern rational updating implies. Table 1 presents the contrast. What makes this a puzzle rather than a difference between two variables is that the two demand measures are not measuring different things. Figure 1 plots the demand attention composite against the activity index: they are positively correlated, they rise together through the mid-2000s expansion, collapse together in 2008–09 and again in 2020, and share every intermediate cycle. Two measures of demand that track each other this closely forecast the same revision in beliefs and opposite movements in prices. Narrative demand and realized demand carry different information, and the question is what in the narrative the market overreacts to.

We answer it by decomposing the narrative into the structural channels through which demand can move prices, including the one the structural literature cannot observe. Following the two-stage procedure of [Kakhbod, Kermani, and Maciel \(2026\)](#), a large language model reads each of 389 monthly *Short-Term Energy Outlook* (STEO) reports published by the EIA between 1983 and 2025. In the first stage the model extracts every analytical claim the EIA offers as a driver of market conditions or of a forecast revision, anchored to a verbatim quotation, with an importance score and the directions the EIA implies for price and for quantity. In the second stage each claim, seen in isolation from its report and its date, is classified into a structural channel. [Kakhbod, Kermani, and Maciel \(2026\)](#) classify a claim about inflation as a demand shock when it implies that prices and output move in the same direction and as a supply shock when they move in opposite directions. We do the same, with the sign pairs of [Kilian \(2009\)](#), who identifies his shocks recursively and characterizes them by their impulse responses: aggregate demand raises price and consumption together, supply moves price and production in opposite directions, and oil-specific demand raises the price with no consumption response but with inventory accumulation, the pattern [Kilian and Murphy \(2014\)](#) subsequently impose as a sign restriction. The classifier matches the price and quantity directions recorded in the first stage to these patterns, and because the two demand channels share a price direction and differ only in which quantity responds, it is also given each channel’s mechanism, aggregate demand operating through the consumption of oil and oil-specific demand through expectations of future scarcity and the incentive to hold oil against them, to separate them. The signs used in classification are the EIA’s stated expectations; whether realized production, consumption, and inventories respond as the categories imply is then a test the paper runs. Aggregating within reports yields five monthly imbalance series, each the net share of the report’s content devoted to one channel, signed so that a positive value denotes expansion of that channel’s force: demand rising, scarcity fears heightening, supply abundant, inventories building, substitution capacity expanding. The oil-specific series, I_{OSD} , reads directly the expectations that [Kilian \(2009\)](#) identifies as the residual price movement orthogonal to supply and activity, and that he describes as “not observable and not linearly related to observables.”

The channels behave as the structural framework says they should where it makes predictions, and the precautionary channel carries the overreaction. Table 3 shows that aggregate-demand content predicts world production growth with the hump-shaped term structure of Kilian’s impulse responses, peaking at twelve months, and predicts U.S. drilling activity from three months onward; precautionary content predicts no production response and a large inventory accumulation, rising to $t = 3.1$ at twelve months, which is the identifying restriction of [Kilian and Murphy \(2014\)](#); inventory content predicts the production cutbacks

that storage theory implies; and supply content predicts production positively. These are the quantity responses the categories were built to satisfy, recovered from text rather than imposed. For returns, aggregate demand and supply content predict returns at no horizon, yet aggregate-demand content predicts strong upward forecast revisions, so realized-demand narratives are updated rationally, as the real economic activity index is. Precautionary content is different. It predicts upward forecast revisions of 0.44 ($t = 4.02$) and lower returns, -2.7% per standard deviation over three months ($t = -3.02$) and -6.0% over twenty-four ($t = -2.20$). The forecast rises and the price falls. The overreaction that the aggregate newspaper measures display is, once demand is decomposed, a property of one component of demand: expectations about future scarcity.

Two further sets of evidence show that the precautionary reversal is a belief that is traded on. First, it forecasts out of sample. In an expanding-window exercise with a ten-year initial training period, the I_{OSD} forecast of the three-month return has an out-of-sample R^2 of 2.2% against the historical mean, with a Clark–West statistic of 1.92 ($p = 0.027$), and is the only channel with positive out-of-sample content (Table 5); the gains arrive in the 2008 collapse, the 2014 glut, and the 2020 shutdown, the episodes in which scarcity narratives were most representative. Second, the futures market records the trade. Using the Commodity Futures Trading Commission’s Commitments of Traders reports, Table 6 shows that a one-standard-deviation increase in precautionary content predicts net selling by non-commercial traders within one to three months, of 0.13 to 0.22 standard deviations of the monthly change (t from -2.6 to -3.1), and by managed-money traders, hedge funds and commodity trading advisors, of 0.33 to 0.36 standard deviations within two to three months ($t = -2.56$ and -2.32). Producers and merchants, the natural counterparty, increase their net long position over the same window ($t = 2.11$ at two months), implying that they hedge less. Speculators are long when the scarcity narrative peaks and leave within a quarter; producers stop selling forward. These effects hold with the prior six-month return in the regression, and the prior return itself has no forecasting power for returns at any horizon, so the reversal is not momentum in disguise. What the prior return does predict is the narrative: a six-month run-up raises next month’s precautionary content by 0.20 standard deviations ($t = 4.64$). The narrative responds to past prices; future prices respond to the narrative.

The overreaction can be located in the price itself. The spot price rises with attention to oil demand and to oil supply, by 8.3% and 4.1% per standard deviation over the preceding three months, and falls by 6.2% and 4.8% over the following six ($t = -3.0$ and -2.3). After precautionary content the spot price falls 4.4% over three months while the futures return is -2.6% : the futures price already sat below the spot price, as the theory of storage implies when a precautionary shock raises the convenience yield, and it anticipated part

of the decline. The ordering after such a shock is spot above futures above the realized price, with analysts' real price forecasts, once deflated by expected inflation, landing near the realized price and not overshooting on average. The whole curve overreacts, the spot most, the futures price less because mean reversion is partly anticipated, and it is prices and the direction of forecast revisions, not the level of analysts' forecasts, that carry the overreaction.

The paper sits at the intersection of three literatures. To the structural decomposition of oil prices (Kilian, 2009; Kilian and Murphy, 2014; Baumeister and Hamilton, 2019; Känzig, 2021) it contributes a direct measurement of the channel that identification by exclusion or sign restriction can only infer, and validates the measurement against the quantity responses that identification implies. To the diagnostic-expectations program (Bordalo, Gennaioli, and Shleifer, 2018; Bordalo, Gennaioli, Ma, and Shleifer, 2020; Bordalo, Gennaioli, La Porta, and Shleifer, 2024) it contributes a market in which the object of belief, future scarcity, is the kind of representative outcome the theory is about, and in which forecast revisions, returns, and positions are all observable, so that the belief, the trade, and the reversal can be seen together. To the literature on speculation in oil (Singleton, 2014; Cheng and Xiong, 2014; Sockin and Xiong, 2015) it contributes a textual state variable that predicts when speculators sell. In method the closest antecedents are Kakhbod, Kermani, and Maciel (2026), whose extraction design we adapt, and Loughran, McDonald, and Pragidis (2019), who document one-day overreaction to an aggregate oil-news tone index; relative to the latter we move from tone to mechanism, from days to the horizons at which forecast errors are priced, and from a single index to a decomposition that says which news is overreacted to. The result resolves the puzzle the newspaper measures pose: aggregate demand attention predicts reversals not because the market misprocesses activity, which it does not, but because demand attention bundles expectations about future scarcity with news about realized consumption, and the market overreacts to the former.

Taken together, the results suggest that expectations about future demand, rather than realized demand or supply, are what drive the oil price movements that subsequently reverse. The structural literature's most important shock is a belief, the belief overshoots, and the overshoot is visible in forecasts, in speculative positions, and in returns. The remainder of the paper proceeds as follows. Section 2 reviews the related literature and Section 3 the data. Section 4 constructs the newspaper attention measures and documents the puzzle. Section 5 decomposes the EIA's outlook into structural channels, validates the channels against quantities, and shows that the overreaction is confined to expectations about future scarcity. Section 6 presents the out-of-sample and positioning evidence and locates the overreaction in the spot price relative to the futures curve and analysts' forecasts. Section 7

reports robustness and Section 8 concludes.

2 Related Literature

A growing body of evidence documents systematic departures from rational expectations in survey-based forecasts of asset prices, macroeconomic aggregates, and firm fundamentals. [Bordalo, Gennaioli, and Shleifer \(2018\)](#) formalize the underlying mechanism as diagnostic expectations: agents overweight outcomes that have become more representative in light of recent data, generating excess volatility, predictable reversals, and overreaction to news. [Bordalo, Gennaioli, Ma, and Shleifer \(2020\)](#) apply the framework across macroeconomic and financial variables using the forecast-revision regression of [Coibion and Gorodnichenko \(2015\)](#), finding that individual forecasters overreact to their own information even when consensus forecasts appear to underreact. [Bordalo, Gennaioli, La Porta, and Shleifer \(2024\)](#) extend it to analysts' long-term earnings growth forecasts and show that overreacting expectations jointly predict forecast errors and returns. [Kohlhas and Walther \(2021\)](#) offer a complementary rationalization based on asymmetric attention to procyclical components of the data. The methodology of comparing forecast revisions to subsequent errors traces to [Nordhaus \(1987\)](#) and [Coibion and Gorodnichenko \(2015\)](#); earlier foundations for under- and overreaction are [Barberis, Shleifer, and Vishny \(1998\)](#) and [Daniel, Hirshleifer, and Subrahmanyam \(1998\)](#), and [Greenwood and Shleifer \(2014\)](#) document the disconnect between survey return expectations and rational benchmarks.

We apply this apparatus to the oil market, where the signature we exploit is a sign reversal between forecast revisions and subsequent realized returns conditional on a textually identified news shock. Our finding that the overreaction is a property of precautionary demand content, and not of aggregate demand or supply content, identifies which news the diagnostic mechanism operates on. It is consistent with the asymmetric-attention reading of [Kohlhas and Walther \(2021\)](#): forecasters engage scarcity narratives through the regime-level reasoning that representativeness theory predicts will trigger overreaction, and process realized activity and production news in a more event-by-event mode that does not.

The structural analysis of oil price fluctuations begins with [Hamilton \(1983\)](#), who documents that oil price increases preceded most postwar U.S. recessions and treats them as exogenous supply disruptions, and [Hamilton \(2003\)](#), who formalizes the supply-shock view. [Kilian \(2009\)](#) reverses the emphasis: he identifies supply, aggregate demand, and oil-specific demand shocks recursively in a monthly VAR and shows that demand shocks dominate the post-1973 sample. [Kilian and Park \(2009\)](#) extend the framework to equity returns. [Kilian and Murphy \(2014\)](#) add above-ground crude inventories and identify a speculative demand

shock through sign restrictions requiring that it raise both the real price and inventories; that identification is the structural counterpart of our precautionary channel, and the inventory results of Table 3 are its textual analogue. [Alquist and Kilian \(2010\)](#) interpret the futures–spot spread as a proxy for precautionary demand under uncertainty about future supply. [Hamilton \(2009\)](#) attributes the 2007–08 spike to demand meeting inelastic supply, [Baumeister and Hamilton \(2019\)](#) revisit identification with Bayesian priors over elasticities and find a larger role for supply, and [Känzig \(2021\)](#) identify supply news shocks from high-frequency futures price changes around OPEC announcements. Our contribution to this literature is a measurement of the precautionary channel that is independent of the identification debate.

A separate strand examines predictability and speculation in oil. [Singleton \(2014\)](#) argues that informational frictions and speculative flows can drive prices away from fundamentals, particularly in the 2008 boom and bust; [Cheng and Xiong \(2014\)](#) survey the financialization literature; [Sockin and Xiong \(2015\)](#) provide a theory in which informational frictions distort the supply–demand signal in commodity prices. The commodity-pricing literature documents momentum, basis, and reversal effects ([Erb and Harvey, 2006](#); [Gorton and Rouwenhorst, 2006](#)), and we show that our textual state variable is not a proxy for past returns. Closest to the quantity side of our evidence, [David \(2019\)](#) shows that exploration activity, a long-run decision that responds to producers’ expectations about future demand, is priced in the risk premium on oil futures; the drilling responses we document in Section 5.5, which differ across the demand channels, are the textual counterpart of that mechanism, and we use his inventory measure as the storage benchmark for the precautionary channel. On forecast accuracy, [Alquist, Kilian, and Vigfusson \(2013\)](#) document that futures and survey forecasts rarely beat the no-change benchmark at short horizons, [Cortazar, Millard, Ortega, and Schwartz \(2019\)](#) find that Bloomberg surveys contain information beyond futures prices, [Garratt, Petrella, and Zhang \(2023\)](#) document systematic biases in EIA forecasts, and [Pierdzioch, Rülke, and Stadtmann \(2010\)](#) document anti-herding among oil forecasters. We differ from these papers in testing for the specific time-series pattern of overreaction rather than for bias in general, and in identifying the shock from text rather than from the forecasts themselves.

Finally, several papers construct news-based series for oil. [Caldara and Iacoviello \(2022\)](#) build a geopolitical risk index from newspapers and relate it to oil prices through the activity channel. [Loughran, McDonald, and Pragidis \(2019\)](#) construct a 130-keyword oil tone index from Dow Jones Energy Service articles and document one-day overreaction. [Kakhbod, Kermani, and Maciel \(2026\)](#) provide the template for our extraction design in a different market. Our newspaper composites follow [Baker, Bloom, and Davis \(2016\)](#) and [Fisher, Martineau, and Sheng \(2022\)](#) in scaling article counts by the volume of the corpus, and our decomposition of the STEO differs from a topic model in that the categories are fixed by

theory before estimation.

3 Data

The paper uses five kinds of data: newspaper text, the EIA’s monthly outlook, futures returns, analyst forecasts, and physical quantities and positions. This section describes their sources and coverage; the construction of the measures built from the two text sources is deferred to Sections 4 and 5, where each is introduced with the question it answers.

Newspapers. We collect articles from TDM Studio through ProQuest, restricting to sources labeled Newspapers or Historical Newspapers and to items classified as News, Article, Feature, or Front page/Cover story, from January 1986 through December 2025. The corpus is used to count articles matching keyword dictionaries for oil demand and oil supply news, scaled by the total number of oil-related articles in each month.

The Short-Term Energy Outlook. The STEO is the EIA’s monthly analytical publication on petroleum and other energy markets. Each report provides forecasts of supply, demand, inventories, prices, and macroeconomic conditions over a twelve- to twenty-four-month horizon, with narrative explanations of the reasoning behind each revision. Reports are released around the eleventh of the month and have been published since 1979. We collect 392 reports covering January 1983 through December 2025, of which 389 are successfully processed; three are excluded for file corruption. Reports were quarterly before 1998 and we carry each forward to the months it covers. The corpus spans every major episode of the past four decades, from the 1986 glut through the Gulf War, the 1998 Asian crisis, the 2000s supercycle, the 2008 crisis, the 2014–2016 shale glut, the 2020 demand collapse, and the 2022 Russia–Ukraine shock.

Futures returns are cumulative log returns on the front-month WTI crude oil futures contract from the end of month t to the end of month $t + h$, for $h = 2, \dots, 12$, from January 1983. The series is the front-month contract from the CME, rolled at expiration, as in [David \(2019\)](#); results using the continuous front-month series from LSEG Datastream are similar. The imbalance for month t is dated to the first trading day after the report’s release, so the predictor is known before the return window opens. Analyst forecasts are the individual responses to the Consensus Economics energy survey of the WTI price at fixed quarter-end targets, from which we take the consensus mean; the revision used in the forecast-revision regressions is the log change in the one-year-ahead consensus between consecutive quarterly surveys. The Survey of Professional Forecasters’ mean forecasts of the GDP deflator, the realized GDP deflator, and the EIA Cushing WTI spot price (from FRED) are used in Section 6.3. Quantities are world crude and total petroleum production and U.S. commercial

crude inventories from the EIA’s international and petroleum statistics, and U.S. rotary rig counts from Baker Hughes as reported by the EIA, each seasonally adjusted.

Positioning is from the CFTC’s Commitments of Traders reports for NYMEX WTI, at month-end. From the legacy report we take the net long position of non-commercial traders, available from January 1986. From the disaggregated report we take the net long position of managed-money traders, which are hedge funds and commodity trading advisors, and of producers, merchants, processors, and users, both available from June 2006. The overlap with the imbalance series is 409 months for non-commercials and 164 months for managed money and producers. Net positions are levels in contracts and trend with open interest, so we work with forward changes over one to twelve months scaled by the trailing thirty-six-month standard deviation of the monthly change, which removes the growth in position variance without using future data. Month-dummy tests on the monthly changes reject at the ten percent level for non-commercials ($p = 0.08$) and managed money ($p = 0.05$) but not for producers ($p = 0.17$), and we seasonally adjust the first two with STL.

Table 2 reports summary statistics for the newspaper composites, the benchmark series, and the analyst forecast variable. The composites are available for 412 months from February 1987 to January 2022, the window over which they overlap the returns and the STEO measures; the real economic activity index of Kilian (2009), constructed from dry-cargo shipping rates, is available from February 1987 to April 2020 and the EIA supply gap from December 1988 to January 2022; the analyst forecast series spans 100 revision cycles from November 1995 to December 2021. Sample sizes differ across tables because each regression uses the longest window over which its particular predictor and outcome overlap; the table notes give the sample for each.

4 The Attention Puzzle

4.1 Measuring attention to oil demand and supply

Our attention measures count articles in historical newspapers that match keyword dictionaries for four directional categories of oil-market news: increases in oil demand, decreases in oil demand, increases in oil supply, and decreases in oil supply. We collect articles from TDM Studio through ProQuest, restricting to sources labeled Newspapers or Historical Newspapers and to items classified as News, Article, Feature, or Front page/Cover story, from January 1986 through December 2025. The dictionaries are derived from the STEO itself. A large language model reads 393 reports and extracts, for each category, the vocabulary that EIA analysts use when describing that kind of development; the vocabulary is supplemented with

forecast-language hooks (*forecast, expect, project, outlook*), product-specific demand terms, and structural macroeconomic framings, with directional verbs combined under proximity operators so that the keywords appear near the relevant directional language. Exclusion clauses strip articles dominated by financial-market language, substitute-fuel discussion, or non-oil demand. Table A2 in the Appendix reports the full queries.

For each calendar month we count articles matching each category and scale by the total number of oil-related articles in the month, following Fisher, Martineau, and Sheng (2022), so that the resulting shares measure the fraction of oil coverage devoted to each direction.¹ The demand composite is the difference between the increasing- and decreasing-demand shares, the supply composite the difference between the increasing- and decreasing-supply shares, and both are standardized over the sample. Two external benchmarks support the interpretation. The demand composite has a monthly correlation of 0.40 with the real economic activity index of Kilian (2009), constructed from dry-cargo shipping rates, rising to 0.43 once both series are smoothed over three months as in the figure, high enough that the two respond to the same global demand conditions and low enough that each retains substantial independent information. The supply composite tracks a realized supply gap constructed from EIA data as the log of U.S. field production plus crude imports excluding the Strategic Petroleum Reserve, less its five-year rolling average, standardized. Figure 1 plots each composite against its benchmark; the demand series jointly capture the mid-2000s supercycle, the 2008–2009 collapse, and the 2020 contraction, and the supply series jointly capture the late-1990s and mid-2010s production expansions and the post-2020 disruptions.

4.2 Empirical design

Our tests follow the diagnostic-expectations framework of Bordalo, Gennaioli, Ma, and Shleifer (2020) and Bordalo, Gennaioli, La Porta, and Shleifer (2024), adapted to a setting in which the news shock is identified from text rather than inferred from the forecasts. The signature of overreaction is a pair of regressions with opposite signs. The forecast-revision regression is

$$\Delta f_{t-j,t+k} = \alpha + \beta \text{News}_t + \varepsilon_t, \quad (1)$$

where Δf is the log change in the consensus forecast between an observation before the news and one after it, with j and k set to span the quarterly revision cycle. The return regression

¹The dictionaries are fixed over the sample. We also constructed a version in which each month’s dictionary was generated only from the reasons in that month’s STEO and applied to articles published before the next report, which is free of look-ahead by construction; the resulting article counts were too sparse to yield usable monthly series. Appendix B reproduces the prompt.

at horizon h is

$$r_{t \rightarrow t+h} = \alpha_h + \beta_h \text{News}_t + \varepsilon_{t,h}, \quad (2)$$

where $r_{t \rightarrow t+h} = \log P_{t+h} - \log P_t$. If $\beta > 0$ in (1) and $\beta_h < 0$ in (2), forecasts move in the direction the news warrants but prices subsequently move against them, which is the pattern that over-extrapolated beliefs unwinding through returns produces; coefficients of the same sign indicate rational updating. We apply the pair to the demand and supply attention composites, with the real economic activity index of Kilian (2009) as the realized-demand benchmark. Every regression is univariate, so that each measure is judged on its own information, and we report ordinary least squares t -statistics, following Kilian (2009).

4.3 Attention predicts reversals; realized demand predicts continuation

Table 1 presents the fact that motivates the paper. Panel A reports (2) at horizons of two to twelve months for the demand attention composite, the supply attention composite, and the real economic activity index, each standardized. Panel B reports (1) for the same three predictors.

Both attention measures predict lower returns, and the effect builds with horizon. Demand attention is uninformative at two to three months, turns negative at four, and is significant from five months onward: -2.5% per standard deviation at five months ($t = -2.39$), -3.1% at six ($t = -2.77$), -4.2% at seven ($t = -3.44$), and -5.7% at twelve ($t = -3.44$). Supply attention follows the same path one month earlier, significant from four months (-2.3% , $t = -2.40$) and reaching -3.8% at six ($t = -3.33$) and -4.5% at twelve ($t = -2.71$). Both also predict upward forecast revisions: 5.6% per standard deviation for demand attention ($t = 5.33$, $R^2 = 0.24$) and 4.1% for supply attention ($t = 2.62$). Forecasters read rising demand and rising supply coverage as bullish and revise upward; prices subsequently fall. That is the signature of overreaction.

The real economic activity index shows what rational updating looks like in the same design. It predicts upward forecast revisions of 4.2% per standard deviation ($t = 6.18$, $R^2 = 0.30$), the same magnitude as the attention measures, and it predicts *higher* subsequent returns, building from 0.8% at two months to 2.2% at six ($t = 1.91$) and 4.3% at twelve ($t = 2.55$). Forecasts and prices move together. Realized activity is news about fundamentals, forecasters incorporate it, and prices confirm it.

The contrast is sharper than a difference between two demand measures, because the two are not measuring different things. Figure 1 plots the demand composite against the activity index and the supply composite against the EIA supply gap. The demand series co-move at

a monthly correlation of 0.40, and 0.43 after three-month smoothing, and share every major cycle: the mid-2000s expansion, the 2008–09 collapse, the 2020 contraction. Yet the one that is a count of newspaper articles predicts a reversal, and the one that is a measure of shipping activity predicts continuation. Two positively correlated measures of demand have opposite forecasting power for returns and the same forecasting power for forecasts.

Panels A and B of Table 3 show that the composites are not merely noise. Demand attention predicts drilling strongly at three and six months (25.2 and 28.1 rigs per standard deviation, $t = 3.25$ and 3.24) and predicts an inventory draw at three months followed by a build at twelve to twenty-four months, the pattern of stocks being run down to meet demand and rebuilt as production responds; supply attention predicts inventory builds from six to thirty-six months and a smaller drilling response. The newspaper measures contain real information about the quantities that adjust to demand and supply conditions. The puzzle is why, given that, they should be systematically overweighted in prices. “Demand” in a newspaper is not a structural object. An article matching the rising-demand dictionary may describe Chinese industrial production, a refinery run increase, a stockpiling program in response to sanctions, or a hurricane, and the composite weights them by column inches. The reversal tells us that something in this bundle is overweighted by the market, and the contrast with the activity index tells us it is not realized consumption. The next section identifies what it is.

One limitation of the attention composites should be stated before we build on them. Their dictionaries are derived from the full corpus of STEO reports, so the vocabulary that identifies rising-demand coverage in 1990 was assembled with knowledge of how the EIA described demand in 2020. This is a form of look-ahead in the construction of the measure, and it is one reason not to treat the composites as tradeable signals. It does not create the puzzle. Analysts revise their forecasts in response to the same coverage, which they observed in real time, and the reversal that follows is in realized prices; a look-ahead vocabulary could make the composites better at identifying demand coverage than a contemporary reader would have been, but it cannot make forecasters revise upward or prices subsequently fall. The decomposition of the next section, moreover, is free of the problem by design, for the reasons [Kakhbod, Kermani, and Maciel \(2026\)](#) give for their own measures: each reason is anchored to a verbatim quotation from a report dated before the outcomes it is used to predict, and the classification stage sees the reason in isolation, without the report, its date, or surrounding context, and is instructed not to draw on knowledge of the episode. The channel measures therefore use only what the EIA wrote at the time, and they are the measures the paper’s conclusions rest on.

5 Decomposing the Puzzle

5.1 Reading the residual

The attention composites bundle every reason a newspaper might write about oil demand, and the structural literature says the reasons matter. Kilian (2009) decomposes real oil price movements into supply, aggregate demand, and a third component he calls oil-specific demand and interprets as precautionary: demand for oil to hold against expected future scarcity. He finds the third component important, but he can only identify it by elimination, as the price innovation orthogonal to production and activity, because, in his words, “expectations shifts related to uncertainty about future oil supplies are not observable and not linearly related to observables.” Kilian and Murphy (2014) pin the same component to inventory behavior through sign restrictions. Neither observes it. If the overreaction in Section 4.3 lives in this component, then decomposing the newspaper bundle requires a measure of precautionary demand that is not a residual.

The STEO provides one. It is the authoritative official analysis of the oil market, institutionally consistent across four decades; it is explicitly analytical, organized around explanations of forecast revisions, so its content is dense in claims about mechanism; and its monthly frequency aligns with the returns, forecasts, and quantities we study. When the EIA writes that “concerns about supply disruptions” or “geopolitical risk” are supporting prices, it is describing precautionary demand in words. We read those words with a language model, classify each claim into the channel of Kilian’s decomposition it describes, and aggregate within reports. The result is five monthly series, one per channel, that measure the net direction of the EIA’s attention to each mechanism: aggregate demand, operating through realized consumption of oil; oil-specific or precautionary demand, operating through expectations of future scarcity and the incentive to hold oil against them; supply, operating through production; inventories, operating through storage; and substitution, operating through long-run displacement of oil. Section 5.3 defines each. The precautionary series is the one the VAR can only infer. The rest of this section describes the extraction, validates the channels against the quantities the structural framework says they should move, and then applies the overreaction test of Section 4.2 to each.

5.2 Extracting structural content from the outlook

Content is extracted in two stages with a large language model (Claude Sonnet 4.5); the prompts are reproduced verbatim in Appendix B. In the first stage the model reads the complete text of each report and returns a structured set of *reasons*, the discrete analytical

claims the EIA offers as drivers of forecast changes or market conditions. Each reason records a short label, a verbatim quotation from the report that supports it, an importance score from zero to ten reflecting the report’s emphasis, the direction of its implied effect on price and on quantity, its horizon, and whether it describes a forecast or a realized condition. A hard inclusion rule requires that every reason be supported by a direct quotation; paraphrase is not permitted, which removes fabrication and over-aggressive summarization, and every reason in the dataset traces to an identifiable sentence in the source. The first stage yields 6,571 reason-rows across the 389 reports, about seventeen per report, a density comparable to the number of paragraph-level claims a typical STEO contains.

In the second stage each reason is classified independently into one of six structural categories, with a signed direction, a one-sentence causal chain recording the classifier’s reasoning, and an indicator for whether the quoted passage explicitly names an oil-market variable. The model sees the reason’s label, its quotation, and the four Stage 1 fields, price direction, quantity direction, horizon, and forecast flag, but not the report, its date, or any other reason from it, and it is instructed not to draw on knowledge of historical episodes outside the passage. The classification procedure follows [Kakhbod, Kermani, and Maciel \(2026\)](#): each category is defined by a mechanism and by the sign pattern its shift implies for price and quantity, and the classifier matches the Stage 1 price and quantity directions to those patterns. Aggregate demand moves price and consumption together; supply moves price and production in opposite directions; oil-specific demand raises the price with consumption flat and inventories accumulating; inventory content is a realized build or draw; and any claim that explicitly invokes a dynamic unfolding over more than a year is assigned to substitution regardless of its short-run signs. Where two categories match the same sign pair, which is the case for the two demand channels, the classifier is instructed to prefer the more specific mechanism, so that a passage about stockpiling in response to sanctions is precautionary and a passage about Chinese industrial production is aggregate demand. The two stages thus separate what was said from what kind of mechanism it implies, and each can be audited on its own. Because the signs used in classification are the EIA’s stated expectations rather than outcomes, the quantity regressions of Section 5.5, which use realized production, consumption, and inventories, remain a test of whether the categories mean what they say.

5.3 Structural categories

The categories operationalize the shocks of [Kilian \(2009\)](#) and [Kilian and Murphy \(2014\)](#), extended with two for content the canonical decomposition does not cover.

Aggregate demand collects reasons that operate through consumption: global GDP and industrial production, transportation activity, OECD and non-OECD consumption forecasts, and macroeconomic conditions affecting realized demand. The defining feature is that the mechanism passes through the burning of oil. *Oil-specific demand* collects reasons that operate through the precautionary or expectational channel: geopolitical tension, sanctions, war risk, anticipated scarcity, OPEC behavior implying future restriction, and any content in which participants respond to expected rather than realized changes in balance. The defining feature is that the mechanism passes through stockpiling; quantity held responds to perceived future risk even if quantity consumed does not. We label the series I_{OSD} to align with Kilian’s taxonomy, but the classification rule is precautionary: it captures the interpretation Kilian gives his residual, and is narrower than the residual itself. *Supply* collects reasons operating through physical production: OPEC quotas, U.S. shale dynamics, offshore activity, decline rates, refinery capacity, and realized or expected extraction. *Inventory* collects reasons operating through storage dynamics that are not better classified as precautionary or as production-driven: commercial stock changes, strategic reserve operations, and seasonal storage. *Substitution-induced* collects reasons operating through long-run substitution: electric vehicles, efficiency, transition policy, biofuel mandates, and structural shifts in oil’s role, whose defining feature is Hotelling-style feedback from expected future demand to current extraction. This category is near-empty before 2007. *Other* is a residual that ensures exhaustive classification and quarantines non-structural content.

5.4 Imbalance series

For each report published in month t and each category c , the imbalance is the net signed share of reasons assigned to that category,

$$I_{c,t} = \frac{1}{|R_t|} \sum_{r \in R_t} \mathbf{1}\{\text{cat}_r = c\} d_r, \quad (3)$$

where R_t is the set of retained reasons in the report, cat_r is the category, and $d_r \in \{+1, -1\}$ is the direction assigned in the second stage, with $+1$ denoting an expansion of the channel’s own force: demand rising for I_{AggDem} , scarcity fears heightening for I_{OSD} , supply abundant for I_{Supply} , inventories building for I_{Inv} , and long-run supply expansion for I_{Sub} , with long-run demand destruction as -1 . Positive values are therefore bullish for prices in the two demand channels and bearish in the supply, inventory, and substitution channels. Retained reasons are those with an importance score of at least five whose quoted passage explicitly names an oil-market variable, and the denominator counts every retained reason in the

report including those assigned to the residual category, so that a value of 0.10 means that expansionary considerations in the channel exceeded contractionary ones by ten percent of the report’s analytical content. The importance score is used to filter and not to weight. Over the 420 months from February 1987 to January 2022 that overlap our returns and quantities, I_{AggDem} has mean 0.103 and standard deviation 0.178, I_{OSD} has mean 0.035 and standard deviation 0.062, I_{Supply} has mean -0.020 and standard deviation 0.185, I_{Inv} has mean 0.006 and standard deviation 0.100, and I_{Sub} has mean -0.021 and standard deviation 0.072. The EIA’s attention to precautionary demand is, on average, tilted toward heightened scarcity fears and is infrequent: the median month has no net precautionary content and the top decile has a share above 0.11.

Before entering any regression, each series is seasonally adjusted with the STL decomposition of [Cleveland, Cleveland, McRae, and Terpenning \(1990\)](#) at monthly periodicity with robust trend estimation, subtracting the estimated seasonal. Several categories have non-trivial seasonal variance shares, most notably inventory (22.7%) and supply (18.8%), reflecting the publication calendar’s winter heating, summer driving, and hurricane season commentary. We do not detrend. A full-sample linear trend explains less than five percent of the variance of any seasonally adjusted series (0.1% for I_{OSD} , with a Newey–West t -statistic of -0.38 on the trend), every series rejects a unit root at the one percent level, and seasonally adjusted and linearly detrended series are correlated above 0.975 for every channel. Section 7 shows that a linear detrend or a Hodrick–Prescott cycle changes nothing. The channel regressions of Sections 5.5 and 5.6 additionally remove the cyclical component of a Hodrick–Prescott filter ($\lambda = 14,400$) and enter the predictors in their natural units, as the table notes state; the seasonally adjusted series are correlated at 0.88 to 0.96 with the HP-cycled ones, and Section 7 shows the results are the same without the filter. Elsewhere predictors are standardized to unit variance so that coefficients are responses to a one-standard-deviation change.

For the channels, inference in the return and quantity regressions uses [Newey and West \(1987\)](#) standard errors with lag truncation equal to the forecast horizon, following [Hodrick \(1992\)](#) and [Boudoukh, Israel, and Richardson \(2022\)](#), because the dependent variable overlaps by construction. The imbalance series have first-order autocorrelations between 0.40 (I_{OSD}) and 0.72 (I_{AggDem}), well below the persistence at which the small-sample bias of [Stambaugh \(1999\)](#) becomes a first-order concern.

Figures 2 and 3 plot them. The top panel of Figure 2 pairs the precautionary series I_{OSD} with the oil-specific demand shock of [Kilian \(2009\)](#), which we recover as the third structural innovation of his three-variable VAR estimated with twenty-four lags on world production, the activity index, and the real refiner acquisition cost of imported crude from 1974 through

2022; the bottom panel pairs the aggregate-demand series I_{AggDem} with the activity index itself. The two precautionary measures share the 1990–91 Gulf War build and collapse, the 2004–08 rise, the 2018–19 run, and the 2020 crash, with a correlation of 0.19 between the text series and the three-month average of the shock. The lead–lag structure is asymmetric: I_{OSD} correlates positively with the shock over the preceding one to five months and negatively with the shock two to three months ahead. The unexplained price rise comes first, the EIA’s explanation shifts toward precautionary considerations, and the precautionary component of the price then gives back part of what it gained. The two measures diverge in 2011–14, when the VAR assigns a large precautionary premium to prices and the text does not; that is the period of the Libyan and Iranian supply shortfalls, which were offset in world production and which the residual therefore books as demand while the EIA described them as supply. The text separates what the residual conflates. Figure 3 plots the supply, inventory, and substitution series; the inventory series tracks the standardized U.S. crude inventory measure of David (2019) at a correlation of 0.67, which fixes its sign convention, positive for builds, and the substitution series is near-zero before 2007.

5.5 The channels recover Kilian’s quantity responses

Before asking which channel is mispriced we ask whether the channels behave as the structural framework says they should where it makes predictions about quantities. Table 3 regresses forward log changes in world crude production and world petroleum consumption, the forward change in U.S. commercial crude inventories, and the forward change in U.S. oil rig counts, over three to thirty-six months, on each measure in turn. The impulse responses of Kilian (2009) give each shock a signature: aggregate demand raises production and consumption; supply moves production and price in opposite directions; oil-specific demand raises the price with no production response and, in the extension of Kilian and Murphy (2014), which imposes the pattern as a sign restriction, with inventory accumulation. The categories were defined to satisfy these signatures, so the test is whether the classifier read the mechanisms correctly.

Aggregate-demand content behaves as realized demand should (Panel C). It predicts production growth at three, six, and twelve months (0.28%, 0.29%, and 0.40% per standard deviation, $t = 1.78$, 2.02, and 2.39), decaying to zero by two years, which is the hump-shaped production response of Kilian’s impulse functions. It predicts drilling from three months (18.9 rigs, $t = 2.93$) through twenty-four ($t = 2.44$), peaking at six months, the investment response of producers to demand they expect to be met. Its inventory response is negative at three and six months ($t = -3.05$ and -1.82) and positive at two to three years

($t = 2.88$ and 2.02): stocks are drawn down to meet consumption and rebuilt once production catches up. The consumption response is flat, which we read as the EIA’s aggregate-demand content being a forecast of activity that production and drilling respond to before realized consumption does.

Precautionary content behaves as Kilian’s residual should (Panel D). It predicts no production response at any horizon ($|t| \leq 1.6$), no consumption response, and no drilling response, and it predicts inventory accumulation that peaks at twelve months (0.24 standard units, $t = 3.14$) and persists at twenty-four ($t = 1.75$). Precautionary demand raises the quantity held without raising the quantity produced or consumed. This is the identifying restriction of Kilian and Murphy (2014), and its appearance here is independent evidence that the text measure captures the channel it was built to capture.

The remaining channels discriminate further. Supply content (Panel E) predicts a drilling contraction at six months (-13.6 rigs, $t = -2.14$) and an inventory build at six months ($t = 2.16$): a report of abundant supply is followed by fuller stocks and less investment, which is the response of producers to a price they expect to be lower. Inventory content (Panel F) is the clearest case of storage dynamics: reports emphasizing builds are followed by production cuts at three and six months (-0.35% and -0.52% , $t = -1.93$ and -2.54) and drilling cuts over the same horizons ($t = -2.28$ and -2.20), by strongly mean-reverting inventories from twelve to thirty-six months ($t = -4.42$, -4.04 , and -2.90), and by a recovery in production and drilling at two years. Substitution content (Panel G) predicts nothing in the full sample, for the reason given above. Across the five channels, the quantity that responds is the quantity the framework says should respond, and, most importantly, the two demand channels are distinguished by exactly the quantity that Kilian’s identification says should distinguish them: aggregate demand moves production, precautionary demand moves inventories.

5.6 Return predictability and forecast revisions by channel

With the channels validated, Table 4 applies the overreaction test of Section 4.2 to each. The first three columns report (2) at three, six, and twelve months; the last two report (1) for revisions to the nearest and to the most distant available forecast horizon. Predictors are in their natural units, so a coefficient is the response to a one-unit change in the net share; Panels C through G of Table 3 report the same return regressions with standardized predictors and at longer horizons.

Two channels are unpriced. Aggregate-demand content predicts returns at no horizon, with coefficients between 0.02 and 0.15 and $|t| \leq 1.1$, and supply content likewise ($|t| \leq 0.6$).

These are the two nulls the interpretation requires. Realized activity and realized production are not forward-looking states; they offer nothing for the representativeness heuristic to act on, and a rational forecaster has no reason to overweight them. What both channels do predict is forecast revisions, and in the direction their content warrants. Aggregate-demand content predicts upward revisions of 0.36 at the short horizon and 0.25 at the long ($t = 4.83$ and 4.66), and supply content predicts downward revisions of 0.11 ($t = -3.16$), a positive supply reading being a report of abundance. Forecasts update and prices confirm: the same-sign pattern the activity index displayed in Table 1, now at the level of the EIA’s own narrative.

The precautionary channel is different. A one-unit increase in I_{OSD} predicts a cumulative return of -0.51 over three months ($t = -3.77$, $R^2 = 3.3\%$) and -0.40 over six ($t = -1.82$); in standardized units, from Table 3, a one-standard-deviation increase predicts -2.7% over three months ($t = -3.02$), -3.4% over twelve ($t = -1.92$), and -6.0% over twenty-four ($t = -2.20$). The same channel predicts *upward* revisions in long-horizon analyst forecasts of 0.44 ($t = 4.02$). The forecast rises and the price falls. Among the five components of the EIA’s narrative, precautionary demand is the only one for which the forecast-revision regression and the return regression have opposite signs. The overreaction that the aggregate newspaper measures display is, once demand is decomposed, a property of expectations about future scarcity.

The inventory channel completes the contrast within the table. It predicts lower returns at three months (-0.13 , $t = -2.44$) and it predicts *downward* forecast revisions (-0.38 and -0.22 , $t = -3.23$ and -3.00). Same sign on both sides: reports emphasizing builds are bearish, forecasters revise down, and prices fall. This is rational updating on storage news. The substitution channel predicts neither returns nor revisions in the full sample, because it has almost no content before 2007. Among the five components of the EIA’s narrative, only expectations about future scarcity display the signature that the newspaper composites display in aggregate. That is the resolution of the puzzle: newspaper attention to demand predicts reversals because it bundles precautionary content with realized-consumption content, and the market overreacts to the former.

5.7 Overreaction or momentum?

The natural alternative to a belief that overshoots is a price that has already moved. If the EIA writes about scarcity after a run-up, and if oil returns mean-revert, I_{OSD} could forecast returns because it marks the top of a price cycle. Adding the prior six-month return to the return regression rules this out. The prior return has no forecasting power for future returns

in any sample or at any horizon, with coefficients that are economically zero and $|t| \leq 0.6$, and its inclusion leaves the precautionary coefficient unchanged: with seasonally adjusted predictors, -1.9% at three months ($t = -2.07$) with and without the control over the full sample, and -3.7% ($t = -2.10$) with and without it after 2007. An interaction between I_{OSD} and the prior return is zero everywhere; the reversal is not concentrated after run-ups.

The narrative, however, is driven by past prices. Regressing $I_{\text{OSD},t}$ on the prior return, controlling for its own lag, the one-month return is uninformative ($t = -0.48$), but the three-month return raises precautionary content by 0.15 standard deviations ($t = 3.46$), the six-month return by 0.20 ($t = 4.64$), and the twelve-month return by 0.21 ($t = 3.99$). The EIA writes more about precautionary demand after prices have risen over the preceding quarter to year. These two results together are the paper’s clearest statement of mechanism. Diagnostic expectations hold that beliefs overweight outcomes that recent data have made representative. A run-up makes future scarcity representative; the agency’s description of the market shifts toward precautionary considerations; and it is this shifted description, not the run-up that produced it, that forecasts the reversal.

6 Out-of-Sample Evidence and Speculative Positioning

6.1 Out-of-sample forecasts

If the precautionary reversal reflects a belief that overshoots, an investor should have been able to exploit it in real time. We test this with the expanding-window design of [Welch and Goyal \(2008\)](#) and [Campbell and Thompson \(2008\)](#). For each channel c and horizon h , we estimate the univariate return regression (2) on an initial window of 120 months and then, at each subsequent month t , re-estimate it on all observations whose h -month return has been realized by t , so that the training sample at origin t ends at $t - h$. The forecast of $r_{t \rightarrow t+h}$ is the fitted value at $I_{c,t}$; the benchmark is the historical mean of the return over the same training sample, which is the forecast a reader who believed returns unpredictable would make. Rolling the origin forward one month at a time produces 297 forecasts from May 1997 to January 2022. We evaluate them with the out-of-sample R^2 of [Campbell and Thompson \(2008\)](#), $R_{\text{OS}}^2 = 1 - \text{MSFE}_{\text{model}}/\text{MSFE}_{\text{mean}}$, which is positive when the model’s squared forecast errors are smaller on average than the benchmark’s, and with the [Clark and West \(2007\)](#) test for nested models, which corrects for the fact that a model with an estimated slope forecasts worse than the mean under the null even when the slope is truly zero. The imbalance series enter seasonally adjusted and HP-filtered, as in Tables 3 and 4, with the filter estimated over the full sample; the regression coefficients, not the filtered

regressor, are what the exercise treats as unknown in real time.

Table 5 reports the results. Panel A compares the six channels at the three-month horizon. Precautionary content is the only channel with positive out-of-sample content: $R_{OS}^2 = 2.2\%$, with a Clark–West statistic of 1.92 ($p = 0.027$). Aggregate demand, supply, and the residual category all forecast worse than the historical mean, with R_{OS}^2 between -0.6% and -1.2% and Clark–West statistics that are negative; inventory and substitution content are indistinguishable from the mean. Panel B traces the precautionary channel across horizons. The gain is concentrated where the in-sample effect is: R_{OS}^2 is 2.2% at three months, 1.9% at four ($p = 0.051$), and 1.1% at five ($p = 0.072$), positive but insignificant at six, and negative at two, seven, and twelve months. An out-of-sample R^2 of two percent is small in absolute terms but is of the same order as the values that the equity-premium literature treats as economically meaningful, and it is obtained here for a single text-based predictor against a benchmark the oil-forecasting literature finds hard to beat (Alquist, Kilian, and Vigfusson, 2013).

The cumulative difference between the benchmark’s and the model’s squared errors, which rises in months when the model forecasts better, shows when the gains accrue. For precautionary content at three months the line is flat from 1997 through 2007, rises in the 2008 collapse, drifts down through 2010–14, rises again in the 2014–15 glut, and makes its largest single move in 2020. The predictive value of precautionary content is therefore episodic: it is earned in the scarcity episodes in which the narrative was most representative and prices subsequently reversed, and it is roughly flat between them. That is the pattern the diagnostic-expectations mechanism implies, since overreaction operates on representative news and not on the average month, but it also means that the out-of-sample result rests on a handful of episodes, and we do not claim more for it than that.

6.2 Precautionary content predicts speculator selling

If beliefs overshoot, someone must trade on them, and the futures market records who. For trader group g we estimate

$$\frac{P_{t+h}^g - P_t^g}{\sigma_t^g} = \alpha_h + \beta_h I_{OSD,t} + \gamma_h r_{t-6 \rightarrow t} + \varepsilon_{t,h}, \quad (4)$$

for $h \in \{1, 2, 3, 6, 9, 12\}$, where P_t^g is the net long position in contracts and σ_t^g is the trailing thirty-six-month standard deviation of the monthly change, so that β_h is measured in units of a typical monthly move. Including the prior six-month return separates selling that follows the narrative from profit-taking that follows a run-up. Table 6 reports the estimates for the

three trader groups.

Precautionary content predicts net selling by financial traders over the following quarter. For non-commercial traders, a one-standard-deviation rise in I_{OSD} predicts a reduction in the net long position of 0.13 standard deviations of the monthly change within one month ($t = -2.86$), 0.21 within two ($t = -3.13$), and 0.22 within three ($t = -2.59$); the effect is not significant beyond three months once the prior return is controlled. For managed-money traders, the reduction is 0.33 within two months ($t = -2.56$), 0.36 within three ($t = -2.32$), and 0.25 within nine ($t = -2.01$). Producers and merchants move the other way, increasing their net long position by 0.30 standard deviations within two months ($t = 2.11$): the commercial side, which is structurally short as a hedge against future output, hedges less when the narrative is most bullish. The prior return predicts positioning as well, and at longer horizons: after a run-up, non-commercials and managed money reduce net length over six to twelve months, with coefficients of -0.41 to -0.83 for non-commercials (t between -3.81 and -4.41) and -0.36 to -0.65 for managed money (t between -2.50 and -3.32), and producers add length at twelve months. The two effects are distinct in horizon. The run-up predicts a slow unwind over the following year; the narrative predicts selling within the quarter, over and above that unwind.

The magnitudes are modest and the sample for managed money is short, 164 months, so we read these results as evidence about the direction of the trade rather than its size. Their significance is in what they rule out. A story in which speculators pile into precautionary narratives would predict buying after high I_{OSD} , and the data show selling. A story in which the EIA merely describes what speculators have already done would predict that positioning subsumes the text's return predictability; on the common post-2006 sample, adding the one-month change in managed-money net length, or its detrended level, to the return regression leaves the I_{OSD} coefficient unchanged at every horizon, and positioning itself is unpriced. The narrative and the positions are two observable traces of the same belief: the belief moves prices, it moves speculators, and each moves back.

6.3 Spot prices, the futures curve, and forecasts of the real price

The return evidence so far concerns the return on a futures position. That leaves open where the overreaction sits: in the spot price, in the futures curve, in analysts' forecasts, or in all three. This section separates them. It also answers a question: how well do the futures curve and analysts forecast the real price of oil, and do they overshoot?

Futures contracts and analyst forecasts are claims on the nominal price, so comparing them to the real price requires an estimate of the inflation expected over the forecast horizon.

We deflate each nominal forecast made at t for a target h months ahead by the expected price level at the target, using the mean forecast of the GDP deflator from the Survey of Professional Forecasters for the survey quarter at the horizon that brackets the target; the realized real price is the EIA Cushing spot price on the first trading day of the target month divided by the realized deflator.² Analyst forecasts come from the Consensus Economics energy survey, which polls between eight and thirty-five institutions on the WTI price at fixed quarter-end targets; the survey was quarterly through 2011, bimonthly through 2015, and monthly thereafter, so we evaluate on a balanced quarterly grid, keeping the survey nearest the middle month of each quarter, which gives 101 surveys and 404 survey–target pairs from August 1995 to January 2022 at horizons of one to thirteen months. For each pair, the futures forecast is the price on the survey date of the contract maturing in the target month, and the no-change forecast is the real spot price on the survey date.

Accuracy and sign. Panel A of Table 7 reports the results. In real terms the futures curve is the most accurate forecast at every horizon, with mean squared errors of 89 to 95 percent of the no-change benchmark; the consensus does not beat no-change at horizons up to nine months and beats it only marginally beyond, in line with [Alquist, Kilian, and Vigfusson \(2013\)](#). Neither forecast overshoots on average. Mean log errors, realized minus forecast, are positive at every horizon; the only one that is significant is the analysts’ at one to three months (0.037, $t = 2.15$), and it says forecasts run low, not high. At the level of individual forecasters, the regression of the real forecast error on the forecaster’s own revision toward the same target over the preceding three months yields a coefficient of -0.19 ($t = -2.30$): analysts overreact to their own revisions, the pattern [Bordalo, Gennaioli, Ma, and Shleifer \(2020\)](#) document for most macroeconomic forecasters, while the consensus coefficient is 0.16 and insignificant, the consensus rigidity of [Coibion and Gorodnichenko \(2015\)](#) (Panel B). Conditional on I_{OSD} at the survey date, the analysts’ real forecast errors are positive and rise with horizon; when precautionary narratives are prominent, analysts adjust their level forecasts too little rather than too much. The analysts, in other words, are not the ones overshooting.

Where the overreaction sits. Table 8 places it. Panel A regresses the h -month change in the spot price and the h -month futures return on I_{OSD} over the same sample. The spot price falls: -3.3% per standard deviation over two months ($t = -3.07$), -4.4% over three ($t = -2.52$), -4.2% over four ($t = -2.53$), and again -5.1% over twelve ($t = -2.92$).

²Each analyst presumably deflates by their own inflation forecast, which we do not observe; using the survey mean for all forecasters and for the futures curve keeps the comparison between them clean at the cost of a small common error.

The futures return is less negative at every horizon, by between one and three percentage points. The difference is the initial basis: the futures price already sat below the spot price when precautionary content was high, so part of the subsequent decline was priced. This is the backwardation that the theory of storage predicts. A precautionary shock raises the convenience yield of holding oil now, which lifts the spot price relative to the futures price and implies an expected decline from that elevated level (Alquist and Kilian, 2010); indeed I_{OSD} predicts backwardation on the survey dates of the Consensus Economics sample with $t = -3.5$. That the front of the curve responds more than the back is the Samuelson effect (Samuelson, 1965; Bessembinder, Coughenour, Seguin, and Smoller, 1995): because fundamentals mean-revert, a shock to today’s price is a smaller shock to the price a year out, and participants who know this price the long end accordingly. The drilling responses of Table 3 are the mechanism at work: rig counts rise within months of demand content and fall when supply is abundant, so production adjusts toward the shock and the price a year out is expected to reflect less of it.

Neither of those observations is evidence of overreaction; both are what a rational market does. The evidence of overreaction is that the futures return itself is negative. Under the theory of storage and the Samuelson effect alone, the futures price is the market’s forecast of the spot price, and its return after the shock would be zero on average. It is not: at three to five months the futures return is -2.6% to -3.0% per standard deviation ($t = -2.1$ to -3.0), and at twelve months -3.5% ($t = -1.97$). The futures price sits above where the spot price ends up. The ordering after a precautionary shock is therefore $S_t > F_t > S_{t+h} \approx$ analysts’ forecast. The whole curve overreacts, the spot most, the futures price less because mean reversion is partly anticipated, and analysts’ level forecasts land near the realized price. The overreaction the paper documents is in prices, and in the direction of forecast revisions, and not in the level of analysts’ forecasts.

Panel B shows the other half of the overshoot, the rise. The spot price increases with attention to oil demand and to oil supply in the same month ($t = 2.5$ and 1.7) and has risen over the preceding three months (8.3% and 4.1% per standard deviation, $t = 4.6$ and 2.5), and it then falls over the following six (-6.2% and -4.8% , $t = -3.0$ and -2.3). Precautionary content is preceded by a run-up (3.1% , $t = 2.9$) and followed by the sharpest reversal (-4.4% at three months). Aggregate-demand content is also preceded by a run-up (4.6% , $t = 2.5$), and the price then stays where it is: no reversal at any horizon. The placebo holds in the spot price as it does in the futures return. That the spot price rises with supply attention, which in our measure is coverage of abundance net of coverage of disruption, suggests that part of the run-up is a response to the volume of oil news rather than to its direction.

7 Robustness

7.1 Filtering

A predictive regression should not depend on how the predictor is filtered, and it does not. We estimate the univariate I_{OSD} return regression under four treatments of the predictor on identical samples: seasonal adjustment only, our baseline; seasonal adjustment followed by a full-sample linear detrend; seasonal adjustment followed by the cyclical component of a two-sided Hodrick–Prescott filter with $\lambda = 14,400$; and the raw imbalance. Over the full sample all four give -1.9% to -2.6% at three months with t between -2.07 and -3.02 ; post-2007 the four treatments give the same coefficient to the second decimal at every horizon, -3.7% at three months, -5.6% to -6.4% at six, and -7.9% to -8.6% at twelve, with t -statistics between -2.1 and -3.8 . The only treatment-dependent feature is that the HP cycle produces additional full-sample significance at eleven and twelve months that the other three do not, and we do not rely on it.

7.2 Inference and persistence

The STEO return regressions use Newey–West standard errors with lag truncation equal to the return horizon; using a fixed truncation of twelve or thirty-six lags instead changes individual t -statistics in both directions but no inference in Table 3 at the five percent level, except that the twelve-month precautionary return coefficient moves between the five and ten percent levels. The first-order autocorrelation of the seasonally adjusted I_{OSD} series is 0.40, so the bias of [Stambaugh \(1999\)](#) is small, and the raw series, with autocorrelation 0.41, gives the same estimates.

7.3 Two regimes

The full-sample estimates are the main result, but the return effect is not uniform over time. In the post-2007 sample of 180 months, the precautionary coefficient grows monotonically with horizon, from -4.4% per standard deviation at three months ($t = -2.40$) to -7.1% at six ($t = -3.55$) and -9.1% at twelve ($t = -3.96$) in a regression with the other four channels; before 2007 the channel predicts a brief reversal at two to three months followed by continuation at seven to ten. The post-2007 market, with its expanded commercial storage, financialized participation, and dense sequence of geopolitical scarcity episodes, is the regime in which precautionary demand shows up as inventory accumulation, as the structural framework implies, and as the sustained return reversal that diagnostic expectations imply. We report the full-sample estimates and note that they average across the two regimes.

7.4 Alternative normalizations

The shared denominator in (3) induces a mechanical negative correlation across categories. Replacing the numerator with its inverse hyperbolic sine, replacing the denominator with an eight-report moving average, and weighting reasons by importance rather than filtering on it each leave the sign and significance of the precautionary coefficients unchanged.

8 Conclusion

Text-based measures of attention to oil demand and supply predict oil futures returns negatively while leading analysts to revise their forecasts upward, the signature of belief over-reaction; the canonical measure of realized demand predicts both in the same direction. Decomposing the analytical content of the EIA’s monthly outlook into the structural channels of Kilian (2009) locates the difference. Aggregate demand and supply content are updated rationally: they move forecasts and leave subsequent returns unaffected, and they move production, inventories, and drilling as the structural framework says they should. Precautionary demand content, the channel that the structural literature identifies as an unobservable expectations shock and that we read directly from the text, moves forecasts up and subsequent returns down, forecasts out of sample, and is followed within a quarter by net selling from speculators and reduced hedging by producers. The narrative responds to past prices, but future prices respond to the narrative. Expectations about future demand, rather than realized demand or supply, are what drive the oil price movements that subsequently reverse.

Several questions remain. Whether the same channel decomposition of newswire text, rather than the official outlook, reproduces the result would establish that the mechanism is a property of the market’s beliefs rather than of one agency’s. Whether the reversal is tradable after transaction costs and roll yield is a question we have not asked. And why the reversal is concentrated after 2007, whether because of the participants who trade on scarcity or the storage that lets precautionary demand express itself in stocks, is an open question that the institutional history of the market may answer.

References

- Alquist, Ron, and Lutz Kilian (2010). What Do We Learn from the Price of Crude Oil Futures? *Journal of Applied Econometrics*, 25(4), 539–573.
- Alquist, Ron, Lutz Kilian, and Robert J. Vigfusson (2013). Forecasting the Price of Oil. In G. Elliott and A. Timmermann (eds.), *Handbook of Economic Forecasting*, Vol. 2A, 427–507. Elsevier.
- Baker, Scott R., Nicholas Bloom, and Steven J. Davis (2016). Measuring Economic Policy Uncertainty. *Quarterly Journal of Economics*, 131(4), 1593–1636.
- Barberis, Nicholas, Andrei Shleifer, and Robert Vishny (1998). A Model of Investor Sentiment. *Journal of Financial Economics*, 49(3), 307–343.
- Baumeister, Christiane, and James D. Hamilton (2019). Structural Interpretation of Vector Autoregressions with Incomplete Identification: Revisiting the Role of Oil Supply and Demand Shocks. *American Economic Review*, 109(5), 1873–1910.
- Bessembinder, Hendrik, Jay F. Coughenour, Paul J. Seguin, and Margaret Monroe Smoller (1995). Mean Reversion in Equilibrium Asset Prices: Evidence from the Futures Term Structure. *Journal of Finance*, 50(1), 361–375.
- Bordalo, Pedro, Nicola Gennaioli, Rafael La Porta, and Andrei Shleifer (2024). Belief Overreaction and Stock Market Puzzles. *Journal of Political Economy*, 132(5), 1450–1484.
- Bordalo, Pedro, Nicola Gennaioli, Yueran Ma, and Andrei Shleifer (2020). Overreaction in Macroeconomic Expectations. *American Economic Review*, 110(9), 2748–2782.
- Bordalo, Pedro, Nicola Gennaioli, and Andrei Shleifer (2018). Diagnostic Expectations and Credit Cycles. *Journal of Finance*, 73(1), 199–227.
- Boudoukh, Jacob, Ronen Israel, and Matthew Richardson (2022). Biases in Long-Horizon Predictive Regressions. *Journal of Financial Economics*, 145(3), 937–953.
- Caldara, Dario, and Matteo Iacoviello (2022). Measuring Geopolitical Risk. *American Economic Review*, 112(4), 1194–1225.
- Campbell, John Y., and Samuel B. Thompson (2008). Predicting Excess Stock Returns Out of Sample: Can Anything Beat the Historical Average? *Review of Financial Studies*, 21(4), 1509–1531.

- Cheng, Ing-Haw, and Wei Xiong (2014). Financialization of Commodity Markets. *Annual Review of Financial Economics*, 6, 419–441.
- Clark, Todd E., and Kenneth D. West (2007). Approximately Normal Tests for Equal Predictive Accuracy in Nested Models. *Journal of Econometrics*, 138(1), 291–311.
- Cleveland, Robert B., William S. Cleveland, Jean E. McRae, and Irma Terpenning (1990). STL: A Seasonal-Trend Decomposition Procedure Based on Loess. *Journal of Official Statistics*, 6(1), 3–73.
- Coibion, Olivier, and Yuriy Gorodnichenko (2015). Information Rigidity and the Expectations Formation Process: A Simple Framework and New Facts. *American Economic Review*, 105(8), 2644–2678.
- Cortazar, Gonzalo, Cristobal Millard, Héctor Ortega, and Eduardo S. Schwartz (2019). Commodity Price Forecasts, Futures Prices, and Pricing Models. *Management Science*, 65(9), 4141–4155.
- Daniel, Kent, David Hirshleifer, and Avanidhar Subrahmanyam (1998). Investor Psychology and Security Market Under- and Overreactions. *Journal of Finance*, 53(6), 1839–1885.
- David, Alexander (2019). Exploration Activity, Long-Run Decisions, and the Risk Premium in Energy Futures. *Review of Financial Studies*, 32(4), 1534–1571.
- Erb, Claude B., and Campbell R. Harvey (2006). The Strategic and Tactical Value of Commodity Futures. *Financial Analysts Journal*, 62(2), 69–97.
- Fisher, Adlai J., Charles Martineau, and Jinfei Sheng (2022). Macroeconomic Attention and Announcement Risk Premia. *Review of Financial Studies*, 35(11), 5057–5093.
- Garratt, Anthony, Ivan Petrella, and Yunyi Zhang (2023). Asymmetry and Interdependence when Evaluating U.S. Energy Information Administration Forecasts. *Energy Economics*, 121, 106620.
- Gorton, Gary, and K. Geert Rouwenhorst (2006). Facts and Fantasies about Commodity Futures. *Financial Analysts Journal*, 62(2), 47–68.
- Greenwood, Robin, and Andrei Shleifer (2014). Expectations of Returns and Expected Returns. *Review of Financial Studies*, 27(3), 714–746.
- Hamilton, James D. (1983). Oil and the Macroeconomy since World War II. *Journal of Political Economy*, 91(2), 228–248.

- Hamilton, James D. (2003). What Is an Oil Shock? *Journal of Econometrics*, 113(2), 363–398.
- Hamilton, James D. (2009). Causes and Consequences of the Oil Shock of 2007–08. *Brookings Papers on Economic Activity*, Spring, 215–261.
- Hodrick, Robert J. (1992). Dividend Yields and Expected Stock Returns: Alternative Procedures for Inference and Measurement. *Review of Financial Studies*, 5(3), 357–386.
- Kakhbod, Ali, Amir Kermani, and Bernardo Maciel (2026). In the Fed’s Mind. *NBER Working Paper No. 35016*, National Bureau of Economic Research.
- Känzig, Diego R. (2021). The Macroeconomic Effects of Oil Supply News: Evidence from OPEC Announcements. *American Economic Review*, 111(4), 1092–1125.
- Kilian, Lutz (2009). Not All Oil Price Shocks Are Alike: Disentangling Demand and Supply Shocks in the Crude Oil Market. *American Economic Review*, 99(3), 1053–1069.
- Kilian, Lutz, and Daniel P. Murphy (2014). The Role of Inventories and Speculative Trading in the Global Market for Crude Oil. *Journal of Applied Econometrics*, 29(3), 454–478.
- Kilian, Lutz, and Cheolbeom Park (2009). The Impact of Oil Price Shocks on the U.S. Stock Market. *International Economic Review*, 50(4), 1267–1287.
- Kohlhas, Alexandre N., and Ansgar Walther (2021). Asymmetric Attention. *American Economic Review*, 111(9), 2879–2925.
- Loughran, Tim, Bill McDonald, and Ioannis Pragidis (2019). Assimilation of Oil News into Prices. *International Review of Financial Analysis*, 63, 105–118.
- Newey, Whitney K., and Kenneth D. West (1987). A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix. *Econometrica*, 55(3), 703–708.
- Nordhaus, William D. (1987). Forecasting Efficiency: Concepts and Applications. *Review of Economics and Statistics*, 69(4), 667–674.
- Pierdzioch, Christian, Jan-Christoph Rülke, and Georg Stadtmann (2010). New Evidence of Anti-Herding of Oil-Price Forecasters. *Energy Economics*, 32(6), 1456–1459.
- Samuelson, Paul A. (1965). Proof That Properly Anticipated Prices Fluctuate Randomly. *Industrial Management Review*, 6(2), 41–49.

- Singleton, Kenneth J. (2014). Investor Flows and the 2008 Boom/Bust in Oil Prices. *Management Science*, 60(2), 300–318.
- Sockin, Michael, and Wei Xiong (2015). Informational Frictions and Commodity Markets. *Journal of Finance*, 70(5), 2063–2098.
- Stambaugh, Robert F. (1999). Predictive Regressions. *Journal of Financial Economics*, 54(3), 375–421.
- Welch, Ivo, and Amit Goyal (2008). A Comprehensive Look at the Empirical Performance of Equity Premium Prediction. *Review of Financial Studies*, 21(4), 1455–1508.

Table 1: Attention to demand and supply, realized demand, forecast revisions, and subsequent returns

<i>Panel A: Cumulative futures return over h months</i>							
	$h = 2$	$h = 3$	$h = 4$	$h = 5$	$h = 6$	$h = 7$	$h = 12$
Demand attention	0.001 (0.23)	−0.003 (−0.39)	−0.014 (−1.47)	−0.025** (−2.39)	−0.031*** (−2.77)	−0.042*** (−3.44)	−0.057*** (−3.44)
Supply attention	0.001 (0.13)	−0.012 (−1.48)	−0.023** (−2.40)	−0.028*** (−2.65)	−0.038*** (−3.33)	−0.043*** (−3.54)	−0.045*** (−2.71)
Real economic activity	0.008 (1.52)	0.012 (1.49)	0.015 (1.56)	0.018* (1.67)	0.022* (1.91)	0.027** (2.19)	0.043** (2.55)
<i>Panel B: Revision in the consensus analyst oil price forecast</i>							
	Coefficient	t		R^2		N	
Demand attention	0.056***	(5.33)		0.238		93	
Supply attention	0.041***	(2.62)		0.070		93	
Real economic activity	0.042***	(6.18)		0.296		93	

Notes: Panel A reports univariate OLS regressions of the form $r_{t \rightarrow t+h} = \alpha_h + \beta_h X_t + \varepsilon_{t,h}$, where $r_{t \rightarrow t+h}$ is the cumulative log return on front-month crude oil futures from the end of month t to the end of month $t+h$, for $h = 2, \dots, 7$ and 12, and X_t is the predictor shown, standardized to unit variance so that β_h is the return response to a one-standard-deviation change. Each cell reports β_h with the OLS t -statistic in parentheses; OLS inference follows the convention of Kilian (2009) for the activity index. The negative return coefficients on the attention composites and the forecast-revision results in Panel B are unchanged in significance under Newey–West standard errors with h lags; the return coefficients on the activity index are not, so the contrast between the two rests on the forecast-revision evidence and on the sign, not the significance, of the activity index’s return predictability. The sample is 364 monthly observations. Panel B reports univariate OLS regressions of $\Delta f_t = \alpha + \beta X_t + \varepsilon_t$, where Δf_t is the log change in the consensus mean analyst forecast of the oil price between consecutive forecast revision cycles and X_t is the same standardized predictor observed before the revision; the sample is 93 revision cycles. Demand attention and supply attention are the newspaper composites of Section 4.1: the share of oil-related articles matching rising-demand keywords less the share matching falling-demand keywords, and likewise for supply, seasonally adjusted and cyclically filtered. Real economic activity is the shipping-based index of global activity from Kilian (2009), entered in levels. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

Table 2: Summary Statistics

Variable	Mean	Std. Dev.	Min	Max	AR(1)	Obs.
<i>Panel A: Unstandardized news-attention composites</i>						
Demand attention	−0.003	1.064	−4.406	3.684	0.802	412
Supply attention	0.017	0.771	−3.487	2.817	0.508	412
<i>Panel B: Standardized news-attention and benchmark variables</i>						
Demand attention	0.000	1.000	−4.139	3.467	0.802	412
Supply attention	0.000	1.000	−4.542	3.629	0.508	412
Kilian AER demand index	0.000	1.000	−2.727	3.145	0.961	391
EIA supply gap	0.000	1.000	−4.049	2.279	0.432	391
<i>Panel C: Analyst forecast variable</i>						
Analyst forecast, log mean	0.013	0.100	−0.363	0.400	0.358	100

Notes: This table reports summary statistics for the news-attention composites, the benchmark oil-market variables, and the analyst forecast variable. Panel A reports the unstandardized demand and supply news-attention composites; Panel B reports standardized versions of the composite and benchmark variables, each demeaned and scaled by its sample standard deviation. AR(1) is the first-order autoregressive coefficient from a regression of each variable on its own one-period lag. The analyst forecast variable is the log consensus mean oil-price forecast. Sample periods are given in Section 4.1.

Table 3: Structural Decomposition of EIA Imbalance Channels: Returns, Quantities, and Forecast Revisions

	Forecast horizon h (months)				
Outcome	$h=3$	$h=6$	$h=12$	$h=24$	$h=36$
<i>Panel A: Demand attention composite</i>					
Returns	−0.0027 (−0.20)	−0.0192 (−1.18)	−0.0660*** (−2.92)	−0.1628*** (−3.25)	−0.0886*** (−2.82)
Production	0.0043 (1.51)	0.0033 (1.30)	0.0021 (0.69)	−0.0086* (−1.81)	−0.0107* (−1.81)
Consumption	−0.0011 (−0.90)	−0.0021 (−0.88)	−0.0053** (−1.98)	−0.0092** (−2.18)	−0.0084* (−1.93)
Inventory	−0.119* (−1.71)	0.010 (0.10)	0.359*** (2.66)	0.666** (2.46)	0.252 (1.63)
Drilling	25.2*** (3.25)	28.1*** (3.24)	3.5 (0.32)	−40.2* (−1.88)	−25.9 (−1.32)
Forecast rev.	0.056*** (5.79)				
<i>Panel B: Supply attention composite</i>					
Returns	−0.0059 (−0.66)	−0.0220 (−1.44)	−0.0387* (−1.77)	−0.0964*** (−3.96)	−0.0769*** (−3.35)
Production	0.0016 (1.34)	0.0011 (0.94)	−0.0010 (−0.57)	−0.0062*** (−2.59)	−0.0061** (−1.97)
Consumption	−0.0007 (−0.96)	−0.0004 (−0.38)	−0.0020 (−1.15)	−0.0040 (−1.60)	−0.0046* (−1.81)
Inventory	−0.003 (−0.06)	0.109 (1.56)	0.345*** (2.97)	0.511*** (3.84)	0.228** (2.21)
Drilling	9.9* (1.71)	10.9 (1.48)	−1.2 (−0.14)	−23.4* (−1.72)	−15.9 (−1.21)
Forecast rev.	0.043* (1.67)				
<i>Panel C: $I_{Aggregate\ Demand}$</i>					
Returns	−0.0006 (−0.07)	0.0107 (0.69)	0.0052 (0.22)	−0.0346 (−0.73)	−0.0617 (−1.43)
Production	0.0028* (1.78)	0.0029** (2.02)	0.0040** (2.39)	0.0019 (0.84)	−0.0041 (−1.58)
Consumption	0.0004 (0.54)	−0.0001 (−0.10)	−0.0014 (−0.89)	−0.0011 (−0.53)	−0.0042* (−1.69)
Inventory	−0.141*** (−3.05)	−0.135* (−1.82)	−0.024 (−0.23)	0.261*** (2.88)	0.211** (2.02)
Drilling	18.9*** (2.93)	26.8*** (3.03)	22.3*** (2.69)	20.4** (2.44)	−3.8 (−0.43)
Forecast rev.	0.245*** (4.66)				

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Table 3 (continued)

Outcome	$h=3$	$h=6$	$h=12$	$h=24$	$h=36$
<i>Panel D: $I_{Oil-Specific Demand}$</i>					
Returns	-0.0265*** (-3.02)	-0.0215* (-1.66)	-0.0343* (-1.92)	-0.0601** (-2.20)	-0.0398* (-1.70)
Production	0.0012 (0.81)	-0.0016 (-0.96)	-0.0021 (-1.09)	-0.0023 (-1.59)	-0.0018 (-0.97)
Consumption	-0.0009 (-0.56)	-0.0015 (-1.31)	-0.0035* (-1.83)	0.0003 (0.23)	-0.0006 (-0.41)
Inventory	0.022 (0.44)	0.084 (1.28)	0.237*** (3.14)	0.170* (1.75)	0.091 (1.16)
Drilling	5.6 (1.29)	2.5 (0.38)	-7.2 (-0.91)	2.7 (0.44)	-0.6 (-0.08)
Forecast rev.	0.443*** (4.02)				
<i>Panel E: I_{Supply}</i>					
Returns	0.0025 (0.34)	-0.0015 (-0.11)	-0.0098 (-0.46)	-0.0144 (-0.46)	-0.0328* (-1.81)
Production	0.0007 (1.23)	-0.0012 (-1.31)	-0.0012 (-0.70)	-0.0004 (-0.27)	-0.0027 (-1.57)
Consumption	-0.0003 (-0.34)	-0.0001 (-0.10)	-0.0002 (-0.15)	-0.0009 (-0.66)	-0.0011 (-0.50)
Inventory	0.061 (1.45)	0.147** (2.16)	0.131 (1.08)	0.062 (0.48)	0.121 (1.19)
Drilling	-2.5 (-0.48)	-13.6** (-2.14)	-19.8* (-1.83)	-20.0 (-1.50)	-16.9 (-1.50)
Forecast rev.	-0.107*** (-3.16)				
<i>Panel F: $I_{Inventory}$</i>					
Returns	-0.0106 (-1.39)	-0.0108 (-0.71)	0.0361 (1.53)	0.0790* (1.79)	0.0548* (1.66)
Production	-0.0035* (-1.93)	-0.0052** (-2.54)	-0.0023 (-0.94)	0.0038 (1.60)	0.0036 (0.88)
Consumption	-0.0007 (-0.90)	0.0012 (1.07)	0.0040** (2.02)	0.0025 (1.50)	0.0028 (1.13)
Inventory	0.041 (0.87)	-0.047 (-0.67)	-0.408*** (-4.42)	-0.602*** (-4.04)	-0.301*** (-2.90)
Drilling	-13.5** (-2.28)	-19.2** (-2.20)	0.4 (0.05)	28.3*** (2.82)	12.2 (1.21)
Forecast rev.	-0.224*** (-3.00)				
<i>Panel G: $I_{Substitution-Induced}$</i>					
Returns	-0.0006 (-0.05)	0.0197 (1.04)	-0.0192 (-0.79)	-0.0137 (-0.34)	-0.0048 (-0.17)

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Table 3 (continued)

Outcome	$h=3$	$h=6$	$h=12$	$h=24$	$h=36$
Production	0.0017 (1.17)	0.0015 (0.87)	-0.0011 (-0.60)	-0.0006 (-0.38)	-0.0009 (-0.35)
Consumption	-0.0003 (-0.32)	-0.0015 (-1.22)	-0.0008 (-0.45)	-0.0023 (-1.14)	0.0001 (0.04)
Inventory	0.028 (0.58)	0.076 (0.78)	0.126 (1.08)	-0.039 (-0.31)	0.008 (0.09)
Drilling	4.9 (0.90)	12.7 (1.38)	11.2 (1.03)	3.0 (0.29)	5.5 (0.52)
<i>Forecast rev.</i>			0.142 (0.51)		

Notes: Each cell reports the slope coefficient from a univariate regression of the indicated outcome on the panel's predictor, with the Newey–West t -statistic in parentheses below, using lag truncation equal to the forecast horizon h . All predictors are seasonally adjusted (STL, period = 12), HP-filtered ($\lambda = 14,400$), and standardized, so coefficients measure the response to a one-standard-deviation movement in the predictor. Outcome units differ: Returns are cumulative log oil futures returns from t to $t + h$; Production and Consumption are forward log changes in seasonally adjusted world crude production and world petroleum consumption; Inventory is the forward change in the standardized David (2019) U.S. crude inventory measure; Drilling is the forward change in HP-cycled U.S. oil rotary rig counts. The *Forecast rev.* row reports a separate quarterly univariate regression ($N = 96$) of the log change in the one-year-ahead consensus analyst forecast between consecutive surveys on the predictor, with Newey–West standard errors (12 lags). Sample periods: returns and drilling 1987:08–2022:01; production 1993:01–2022:01; consumption and inventory 1990:01–2022:01. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Univariate Return Predictability and Forecast Revisions by EIA Imbalance Category

	Forward log oil return at horizon h			Analyst forecast revisions [†]	
	$h = 3$	$h = 6$	$h = 12$	Short-run	Long-run
$I_{\text{Aggregate Demand}}$	0.024 (0.55) [0.000]	0.112 (1.08) [0.004]	0.145 (1.09) [0.003]	0.355*** (4.83) [0.148]	0.245*** (4.66) [0.111]
$I_{\text{Oil-Specific Demand}}$	-0.507*** (-3.77) [0.033]	-0.401* (-1.82) [0.010]	-0.417 (-1.09) [0.005]	0.295 (1.56) [0.014]	0.443*** (4.02) [0.053]
I_{Supply}	0.010 (0.15) [0.000]	-0.007 (-0.06) [0.000]	-0.117 (-0.63) [0.002]	-0.113* (-1.65) [0.015]	-0.107*** (-3.16) [0.025]
$I_{\text{Inventory}}$	-0.129** (-2.44) [0.006]	-0.140 (-1.11) [0.003]	0.269 (1.16) [0.005]	-0.377*** (-3.23) [0.059]	-0.224*** (-3.00) [0.036]
$I_{\text{Substitution-Induced}}$	0.009 (0.03) [0.000]	0.383 (0.75) [0.009]	-0.476 (-0.98) [0.006]	0.002 (0.01) [0.000]	0.142 (0.51) [0.005]
I_{Other}	0.054 (0.33) [0.000]	0.248 (0.72) [0.002]	-0.224 (-0.69) [0.001]	0.570 (1.31) [0.017]	0.315 (0.98) [0.011]
N	402			96	

Notes: Each cell reports three statistics from a univariate regression of the indicated dependent variable on the row predictor: the slope coefficient (top), the t - or z -statistic in parentheses (middle), and the regression R^2 in square brackets (bottom). Columns $h = 3, 6, 12$ use cumulative log oil futures returns from t to $t + h$ as the dependent variable, with Newey-West HAC standard errors using 36 lags. All return predictors are seasonally adjusted (STL, period = 12, robust) and HP-filtered ($\lambda = 14400$) prior to estimation. Returns sample: 1987:08–2021:01. [†] The Forecast Revisions columns report regressions of the log mean analyst forecast revision on each row predictor individually. The Short-run column uses revisions to forecasts for the nearest available horizon following each EIA STEO report; the Long-run column uses revisions to forecasts for the most distant available horizon. Sample frequency is quarterly with $N = 96$. Both columns use Newey-West HAC standard errors with 12 lags. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Out-of-Sample Return Forecasts by Imbalance Channel

<i>Panel A: Three-month horizon, by channel</i>				
	R_{OS}^2 (%)	Clark–West	p -value	N
$I_{\text{Aggregate Demand}}$	−1.24	−1.82	0.965	297
$I_{\text{Oil-Specific Demand}}$	2.23	1.92	0.027**	297
I_{Supply}	−0.58	−2.22	0.987	297
$I_{\text{Inventory}}$	−0.17	0.62	0.269	297
$I_{\text{Substitution-Induced}}$	−1.10	0.14	0.446	297
I_{Other}	−0.63	−1.28	0.900	297
<i>Panel B: $I_{\text{Oil-Specific Demand}}$, by horizon</i>				
	R_{OS}^2 (%)	Clark–West	p -value	N
$h = 2$	−0.87	0.34	0.367	298
$h = 3$	2.23	1.92	0.027**	297
$h = 4$	1.85	1.64	0.051*	296
$h = 5$	1.07	1.46	0.072*	295
$h = 6$	0.43	1.02	0.154	294
$h = 7$	−1.20	−0.54	0.706	293
$h = 12$	−0.19	−0.32	0.624	288

Notes: Each row reports an expanding-window out-of-sample forecast of the cumulative log oil futures return over h months from a univariate regression on the indicated imbalance series, against the historical-mean benchmark. The initial estimation window is 120 months; at each forecast origin t the regression is re-estimated on all observations whose h -month return is realized by t , and the benchmark is the mean return over the same observations. Forecasts run from May 1997 to January 2022. R_{OS}^2 is the [Campbell and Thompson \(2008\)](#) out-of-sample R^2 , $1 - \text{MSFE}_{\text{model}}/\text{MSFE}_{\text{mean}}$, in percent. Clark–West is the [Clark and West \(2007\)](#) adjusted-MSFE statistic for nested models, with the one-sided p -value from the standard normal distribution; the variance of the loss differential is not adjusted for the overlap of h -month forecast errors, so p -values at longer horizons are understated. Predictors are seasonally adjusted (STL, period = 12) and HP-filtered ($\lambda = 14,400$). N is the number of out-of-sample forecasts; in Panel B it declines with h because the first forecast origin is $120 + h$ months into the sample. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Precautionary content and speculative positioning

	$h = 1$	$h = 2$	$h = 3$	$h = 6$	$h = 9$	$h = 12$
<i>Panel A: Non-commercial traders, legacy report ($N = 409$)</i>						
I_{OSD}	-0.127*** (-2.86)	-0.212*** (-3.13)	-0.219*** (-2.59)	-0.078 (-0.89)	-0.027 (-0.28)	-0.090 (-0.74)
$r_{t-6 \rightarrow t}$	-0.081 (-1.59)	-0.132* (-1.92)	-0.155* (-1.96)	-0.407*** (-3.81)	-0.575*** (-3.50)	-0.825*** (-4.41)
<i>Panel B: Managed money, disaggregated report ($N = 164$)</i>						
I_{OSD}	-0.090 (-1.04)	-0.331** (-2.56)	-0.355** (-2.32)	-0.133 (-1.12)	-0.247** (-2.01)	-0.133 (-0.71)
$r_{t-6 \rightarrow t}$	-0.123* (-1.80)	-0.211** (-2.14)	-0.171 (-1.57)	-0.360** (-2.50)	-0.651*** (-3.32)	-0.627** (-2.54)
<i>Panel C: Producers, merchants, processors, and users ($N = 164$)</i>						
I_{OSD}	0.131 (1.41)	0.295** (2.11)	0.282 (1.58)	0.041 (0.14)	0.164 (0.50)	0.165 (0.36)
$r_{t-6 \rightarrow t}$	0.152* (1.89)	0.155 (1.23)	0.137 (0.80)	0.140 (0.53)	0.458 (1.37)	0.822** (1.98)

Notes: Dependent variable is the change in the group's net long position in NYMEX WTI futures from month t to $t+h$, divided by the trailing 36-month standard deviation of the monthly change (equation (4)); non-commercial and managed-money changes are STL seasonally adjusted. I_{OSD} is STL adjusted and standardized; $r_{t-6 \rightarrow t}$ is the standardized prior six-month log return. Newey–West t -statistics with h lags in parentheses. Panel A from January 1987; Panels B and C from June 2006.

Table 7: Real Oil Price Forecasts: Accuracy, Sign, and Overreaction

<i>Panel A: Forecast accuracy and mean error by horizon, quarterly grid (log real price)</i>					
Horizon (months)	Forecast	RMSE	MSFE ratio	Mean error	N
1–3	Futures	0.144	0.947	0.007 (0.45)	101
	Analysts	0.159	1.150	0.037 (2.15)	101
	No change	0.148	1.000	0.009 (0.57)	101
4–6	Futures	0.239	0.930	0.020 (0.69)	101
	Analysts	0.254	1.056	0.051 (1.61)	101
	No change	0.248	1.000	0.017 (0.64)	101
7–9	Futures	0.283	0.954	0.036 (0.88)	101
	Analysts	0.300	1.073	0.058 (1.28)	101
	No change	0.290	1.000	0.026 (0.65)	101
10–13	Futures	0.319	0.890	0.052 (1.00)	101
	Analysts	0.335	0.987	0.064 (1.12)	101
	No change	0.338	1.000	0.031 (0.60)	101
<i>Panel B: Forecast error on forecast revision, $FE_{i,t} = a + b REV_{i,t} + \varepsilon$</i>					
	b	t	N	Forecasters	
Individual forecasters, all surveys	−0.189	(−2.30)	8,963	75	
Consensus mean, quarterly grid	0.159	(1.27)	291	—	

Notes: Forecasts are from the Consensus Economics energy survey of WTI crude oil with fixed quarter-end targets, August 1995 to January 2022. Because the survey moved from quarterly to monthly over the sample, Panel A uses a balanced quarterly grid, keeping the survey nearest the middle month of each quarter (101 surveys, 404 survey–target pairs). Each nominal forecast made at survey date t for target $t + h$ is deflated by the survey-quarter mean SPF forecast of the GDP deflator at the horizon bracketing the target; the realized real price is the EIA Cushing WTI spot price on the first trading day of the target month divided by the realized GDP deflator. The futures forecast is the survey-date price of the contract maturing in the target month; the no-change forecast is the real spot price on the survey date. RMSE is the root mean squared log error; MSFE ratio is the ratio of mean squared error to that of the no-change forecast; mean error is the mean log error (realized minus forecast) with a Newey–West t -statistic in parentheses, lags equal to the horizon in quarters. On all 156 surveys the MSFE ratios are 0.79–0.88 for futures and 0.85–0.98 for analysts, and no mean error is significant. Panel B regresses the real log forecast error on the revision of the same forecast toward the same target; for individual forecasters the revision is relative to the forecaster’s most recent prior forecast at least three months earlier, with standard errors clustered by survey; for the consensus it is between consecutive grid surveys, with Newey–West standard errors. $b < 0$ indicates overreaction and $b > 0$ underreaction in the sense of Coibion and Gorodnichenko (2015) and Bordalo, Gennaioli, Ma, and Shleifer (2020).

Table 8: Spot Price, Futures Price, and Precautionary Content

<i>Panel A: Spot return and futures return on I_{OSD}, by horizon</i>				
h (months)	Spot $\log(S_{t+h}/S_t)$	Futures $\log(S_{t+h}/F_t)$	Difference	N
2	−0.033 (−3.07)	−0.007 (−1.48)	0.026	420
3	−0.044 (−2.52)	−0.026 (−3.02)	0.017	420
4	−0.042 (−2.53)	−0.030 (−2.40)	0.012	420
5	−0.037 (−2.20)	−0.027 (−2.13)	0.010	420
6	−0.026 (−1.31)	−0.021 (−1.59)	0.005	420
9	−0.037 (−1.91)	−0.004 (−0.21)	0.033	420
12	−0.051 (−2.92)	−0.035 (−1.97)	0.016	410
<i>Panel B: Spot price around attention and content, contemporaneous and forward</i>				
	Same month	Prior 3 months	Next 3 months	Next 6 months
Demand attention	0.022 (2.49)	0.083 (4.59)	−0.029 (−1.67)	−0.062 (−3.03)
Supply attention	0.011 (1.72)	0.041 (2.53)	−0.022 (−1.82)	−0.048 (−2.33)
I_{OSD}	0.005 (0.69)	0.031 (2.87)	−0.044 (−2.52)	−0.026 (−1.31)
I_{AggDem}	0.009 (1.23)	0.046 (2.47)	−0.006 (−0.43)	−0.005 (−0.25)

Notes: The spot price is the EIA Cushing WTI price at the end of each month (FRED series DCOILWTICO); the futures return is the front-month contract return used throughout the paper. Predictors are STL seasonally adjusted, HP-filtered, and standardized, as in Table 3. Panel A reports coefficients from univariate regressions of the h -month log change in the spot price and the h -month futures return on I_{OSD} over the same observations, February 1987 to January 2022, with Newey–West t -statistics (h lags) in parentheses. Difference is the futures coefficient minus the spot coefficient; a positive value means the futures price sat below the spot price when I_{OSD} was high and anticipated part of the subsequent decline. Panel B reports coefficients from regressions of the log change in the spot price over the same month, the preceding three months, the following three months, and the following six months on each predictor, with Newey–West t -statistics (lags equal to the window). The same-month and prior-window columns describe co-movement, not prediction.

Text-Based Oil Demand and Supply Measures

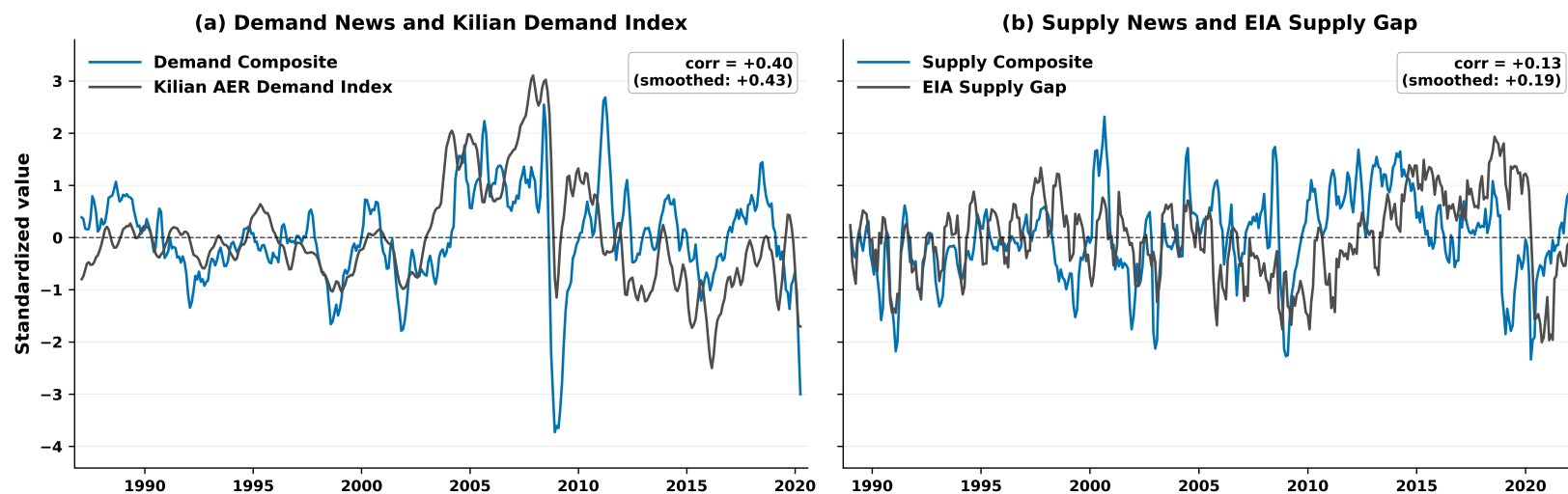


Figure 1: Text-Based Oil Demand and Supply Measures. This figure compares the text-based oil demand and supply composites to external benchmark measures. Panel (a) plots the demand composite against the Kilian AER demand index. Panel (b) plots the supply composite against the EIA supply gap. The EIA supply gap is defined as the deviation of log U.S. crude supply from a five-year rolling trend, where crude supply is U.S. field production plus crude imports excluding the Strategic Petroleum Reserve. All series are converted to monthly frequency, standardized to have mean zero and unit variance, and smoothed using a rolling average.

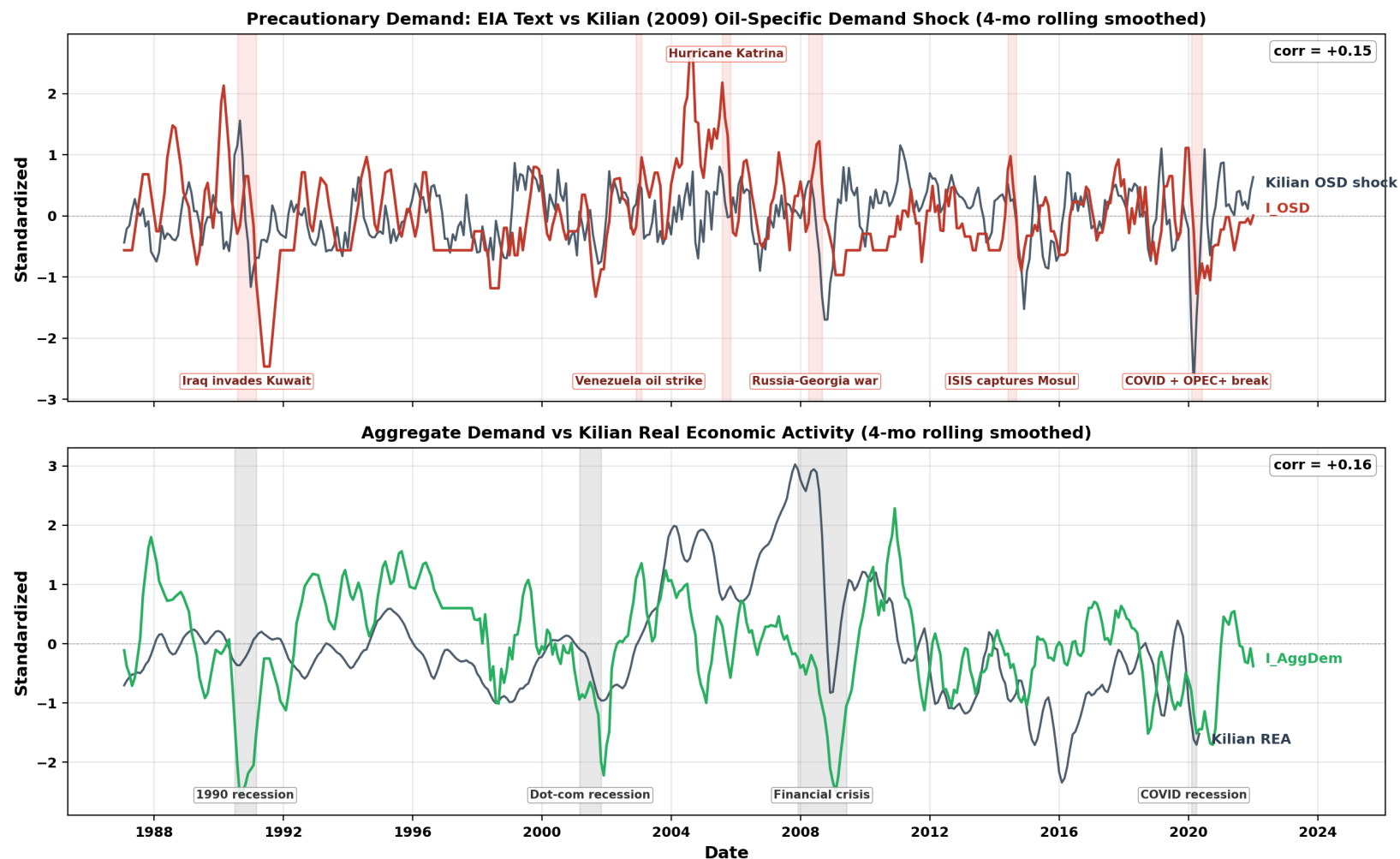


Figure 2: Demand Imbalances from the EIA Outlook and Their Structural Counterparts. The top panel plots the oil-specific (precautionary) demand imbalance I_{OSD} extracted from the *Short-Term Energy Outlook* against the oil-specific demand shock of Kilian (2009), recovered as the third structural innovation from a three-variable VAR in world crude production, the real economic activity index, and the real refiner acquisition cost of imported crude, with twenty-four lags and recursive identification. The bottom panel plots the aggregate-demand imbalance I_{AggDem} against the real economic activity index itself. All series are standardized and smoothed with a centered four-month moving average; the correlation between the smoothed series appears in the upper right of each panel. Red shading in the top panel marks major geopolitical and supply-disruption episodes; gray shading in the bottom panel marks NBER recessions. The figure pairs each text-based demand measure with the structural measure of the same channel: the top panel compares two measures of precautionary demand, the bottom panel two measures of realized demand.

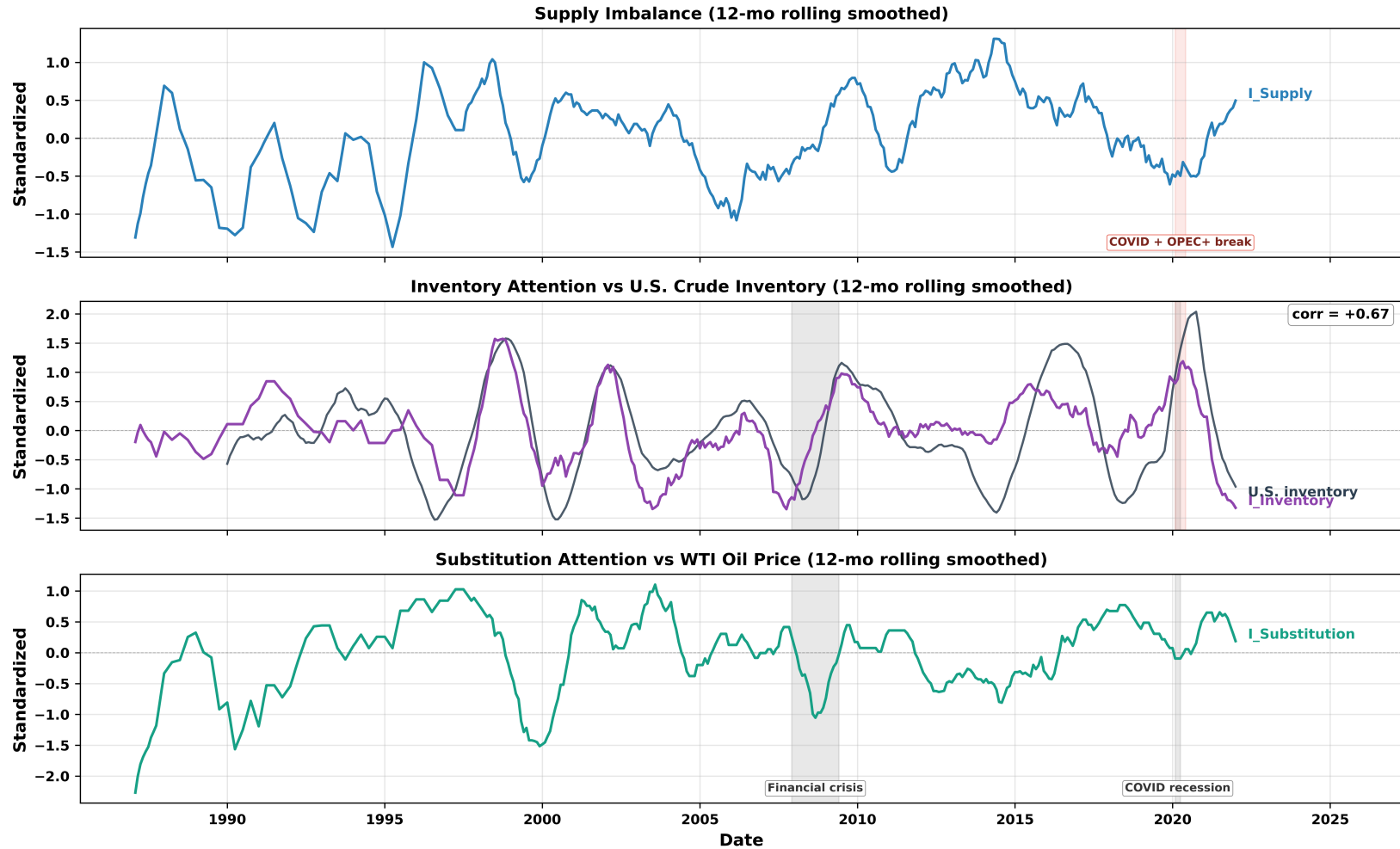


Figure 3: Supply-Side Imbalances from the EIA Outlook. The top panel plots the supply imbalance I_{Supply} extracted from the *Short-Term Energy Outlook*, where a positive value indicates that the report's supply-related content is net toward abundance (production increases, capacity additions, or outages resolving) rather than restraint or disruption. The middle panel plots the inventory imbalance I_{Inv} , where a positive value indicates net emphasis on inventory builds, against the standardized U.S. commercial crude inventory measure of David (2019); the correlation between the smoothed series is 0.67. The bottom panel plots the substitution-induced imbalance I_{Sub} , where a positive value indicates net content on long-run supply expansion and a negative value net content on long-run demand destruction, including substitution away from oil, against the real WTI price. All series are standardized and smoothed with a centered twelve-month moving average. Gray shading marks the 2007–09 and 2020 NBER recessions; red shading marks the 2020 demand collapse and OPEC+ breakdown. The inventory series tracks the realized stock cycle closely, including the 2008–09 and 2015–16 builds and the 2020 spike; the substitution series is sparse before 2007 and thereafter co-moves with the price level, rising in the high-price episodes of 2011–14 and 2018 that prompt discussion of alternatives.

A Search Terms

Table [A1](#) summarizes the keyword groups that define the four directional attention categories of Section [4.1](#), and Table [A2](#) reproduces the four TDM Studio queries exactly as they were run.

Table A1: Text Search Terms for Oil Demand and Supply News

Category	Oil-market terms	Search-word groups
<i>Panel A. Demand and supply search categories</i>		
Supply increase	oil; “crude oil”; petroleum; “liquid fuels”	production; output; supply; “crude oil production”; increase*; rise*; grow*; ramp*; boost*; “come online”; “back online”; OPEC; “OPEC+”; quota*; “production target*”; “production ceiling”; raise*; add*; resume*; restart*; restore*; recover*; export*; terminal*; port*; “spare capacity”; “surplus capacity”; expand*; higher; inventor*; “commercial stock*”; build*; “stock build”.
Supply decrease	oil; “crude oil”; petroleum; “liquid fuels”	production; output; supply; “crude oil production”; cut*; reduc*; curtail*; restrain*; declin*; fall*; drop*; “shut in”; shut-in*; OPEC; “OPEC+”; quota*; “production target*”; “production agreement”; extend*; “voluntary reduction*”; compliance; over-compliance; outage*; disruption*; “force majeure”; shutdown*; export*; pipeline; terminal*; port*; loadings; shipment*; halt*; suspend*; stop*; inventor*; “commercial stock*”; “days-of-supply”; draw*; withdraw*; drawdown; low; tight; “below normal”; “spare capacity”; “surplus capacity”; limited; dwindle*; “little cushion”.
Demand increase	oil; “crude oil”; petroleum; “liquid fuels”; “petroleum products”	demand; consumption; “product supplied”; increase*; rise*; grow*; rebound*; recover*; strengthen*; forecast*; expect*; project*; outlook; “motor gasoline”; gasoline; “jet fuel”; distillate; diesel; “refinery run*”; throughput; utilization; “crude input*”; “gross input*”; higher; record*.
Demand decrease	oil; “crude oil”; petroleum; “liquid fuels”; “petroleum products”	demand; consumption; “product supplied”; declin*; fall*; drop*; weaken*; slump*; contract*; “revised down”; “demand growth”; “consumption growth”; slow*; lower; moderat*; forecast*; expect*; project*; outlook; “motor gasoline”; gasoline; “jet fuel”; distillate; diesel.
<i>Panel B. Excluded terms</i>		
Supply exclusions	WTI; Brent; futures; option*; volatility; “spot price”; “\$/bbl”; “per barrel”; stock*; equity; ETF*; bond*; yield*; “supply chain”; “labor supply”; LNG; “natural gas”; coal.	
Demand exclusions	WTI; Brent; futures; option*; volatility; backwardation; contango; “spot price”; “price forecast”; “\$/bbl”; “per barrel”; stock*; equity; ETF*; bond*; yield*; “supply chain”; “labor demand”; “electricity demand”; LNG; “natural gas”; coal.	

Notes: This table summarizes the high-precision text searches used to construct the oil demand and supply news measures. Searches are applied to oil-related articles from January 1986 to December 2025. The truncation character (*) denotes all possible endings of a word. The supply composite is defined as supply-increase news minus supply-decrease news. The demand composite is defined as demand-increase news minus demand-decrease news.

Table A2: TDM Studio Search Queries for Oil Demand and Supply News (as run)

Category	Query
Demand increase	((oil OR "crude oil" OR petroleum OR "liquid fuels" OR "petroleum products") AND (((demand OR consumption OR "product supplied") NEAR/5 (increase* OR rise* OR grow* OR rebound* OR recover* OR strengthen*)) OR ((forecast* OR expect* OR project* OR outlook) NEAR/5 (demand OR consumption) NEAR/5 (increase* OR rise* OR grow*)) OR (("motor gasoline" OR gasoline OR "jet fuel" OR distillate OR diesel) NEAR/5 (demand OR consumption OR "product supplied") NEAR/5 (increase* OR rise* OR grow*)) OR (("refinery run*" OR throughput OR utilization OR "crude input*" OR "gross input*") NEAR/5 (increase* OR rise* OR higher OR record*)))) NOT (WTI OR Brent OR futures OR option* OR volatility OR backwardation OR contango OR "spot price" OR "price forecast" OR "\$/bbl" OR "per barrel" OR stock* OR equity OR ETF* OR bond* OR yield* OR "supply chain" OR "labor demand" OR "electricity demand" OR LNG OR "natural gas" OR coal)
Demand decrease	((oil OR "crude oil" OR petroleum OR "liquid fuels" OR "petroleum products") AND (((demand OR consumption OR "product supplied") NEAR/5 (declin* OR fall* OR drop* OR weaken* OR slump* OR contract* OR "revised down")) OR (("demand growth" OR "consumption growth") NEAR/5 (slow* OR weaken* OR lower OR moderat*)) OR ((forecast* OR expect* OR project* OR outlook) NEAR/5 (demand OR consumption) NEAR/5 (declin* OR lower OR "revised down")) OR (("motor gasoline" OR gasoline OR "jet fuel" OR distillate OR diesel) NEAR/5 (demand OR consumption OR "product supplied") NEAR/5 (declin* OR fall* OR drop* OR weaken*)))) NOT (WTI OR Brent OR futures OR option* OR volatility OR backwardation OR contango OR "spot price" OR "price forecast" OR "\$/bbl" OR "per barrel" OR stock* OR equity OR ETF* OR bond* OR yield* OR "supply chain" OR "labor demand" OR "electricity demand" OR LNG OR "natural gas" OR coal)
Supply increase	((oil OR "crude oil" OR petroleum OR "liquid fuels") AND (((production OR output OR supply OR "crude oil production") NEAR/5 (increase* OR rise* OR grow* OR ramp* OR boost* OR "come online" OR "back online")) OR ((OPEC OR "OPEC+" OR quota* OR "production target*" OR "production ceiling") NEAR/8 (increase* OR raise* OR boost* OR add*)) OR ((resume* OR restart* OR restore* OR recover*) NEAR/6 (production OR output OR export* OR terminal* OR port*)) OR (("spare capacity" OR "surplus capacity") NEAR/6 (increase* OR expand* OR higher)) OR ((inventor* OR "commercial stock*") NEAR/5 (build* OR increase* OR rise* OR "stock build")))) NOT (WTI OR Brent OR futures OR option* OR volatility OR "spot price" OR "\$/bbl" OR "per barrel" OR stock* OR equity OR ETF* OR bond* OR yield* OR "supply chain" OR "labor supply" OR LNG OR "natural gas" OR coal)
Supply decrease	((oil OR "crude oil" OR petroleum OR "liquid fuels") AND (((production OR output OR supply OR "crude oil production") NEAR/5 (cut* OR reduc* OR curtail* OR restrain* OR declin* OR fall* OR drop* OR "shut in" OR shut-in*)) OR ((OPEC OR "OPEC+" OR quota* OR "production target*" OR "production agreement") NEAR/8 (cut* OR reduc* OR extend* OR "voluntary reduction*" OR compliance OR over-compliance)) OR ((outage* OR disruption* OR "force majeure" OR shutdown*) NEAR/6 (production OR output OR export* OR pipeline OR terminal* OR port*)) OR ((exports OR loadings OR shipment*) NEAR/6 (halt* OR suspend* OR disrupt* OR stop* OR declin*)) OR ((inventor* OR "commercial stock*" OR "days-of-supply") NEAR/5 (draw* OR withdraw* OR drawdown OR declin* OR low OR tight OR "below normal")) OR (("spare capacity" OR "surplus capacity") NEAR/6 (low OR limited OR dwindle* OR declin* OR "little cushion")))) NOT (WTI OR Brent OR futures OR option* OR volatility OR "spot price" OR "\$/bbl" OR "per barrel" OR stock* OR equity OR ETF* OR bond* OR yield* OR "supply chain" OR LNG OR "natural gas" OR coal)

Notes: Each query is run in TDM Studio (ProQuest) over January 1986 to December 2025, restricted to sources labeled Newspapers or Historical Newspapers and to items classified as News, Article, Feature, or Front page/Cover story. The truncation character (*) matches all endings of a word; NEAR/*n* requires the bracketed terms to appear within *n* words of each other. The NOT clause removes articles dominated by financial-market language, non-oil demand or supply, and other fuels. The demand composite is the monthly share of oil-related articles matching the demand-increase query less the share matching the demand-decrease query; the supply composite is defined analogously.

B Extraction Prompts and Imbalance Construction

This appendix reproduces the system prompts used in the two-stage extraction of Section 5.2 and documents the construction of the imbalance series of Section 5.4. Both stages call Claude Sonnet 4.5 through the Anthropic API. Reports are read from PDF with front matter, tables of contents, data tables, and pages with unreadable text removed before extraction. In the first stage the model receives the full report text as the user message, with the system prompt below, and returns a JSON list of reasons. In the second stage each reason is sent separately as a user message containing its label, quotation, and the four Stage 1 fields (price direction, quantity direction, horizon, and forecast flag), with the system prompt below, and the model returns a category, a one-sentence causal chain, an explicit-reference flag, and a signed direction. Long documents are not truncated. The Stage 1 response is prefilled with an opening brace to force JSON output; malformed responses are repaired by a JSON parser and, on failure, logged and re-queried.

B.1 Stage 1 system prompt: reason extraction

You are an energy economist analyzing an Energy Information Administration Short-Term Energy Outlook (STEO) report. Your task is to identify all reasons that the report explicitly cites as structural causes or drivers of OIL-MARKET conditions specifically -- including current or expected movements in crude oil prices, oil supply, oil demand, oil inventories, or refined petroleum product markets.

The STEO covers multiple energy commodities (oil, natural gas, coal, electricity, renewables). Your task is OIL-SPECIFIC. You must extract only reasons that drive the oil market, not reasons that drive natural gas, coal, electricity, or other energy commodities -- unless those commodities explicitly affect oil through substitution.

Use the structural rules below.

=====

HARD INCLUSION RULES

=====

Rule 1 (Oil-market trigger): A reason may be included only if the quoted sentence or passage in your "explanation" field contains an explicit reference to one or more of the following oil-market terms:

ALLOWED OIL-MARKET TERMS (the passage must contain at least one):

- crude oil, crude price, crude production, crude supply, crude demand, crude inventory
- oil price, oil production, oil supply, oil demand, oil consumption, oil inventory
- oil imports, oil exports, oil stocks, OECD oil stocks, OECD petroleum stocks
- petroleum, petroleum products, petroleum supply, petroleum inventory
- gasoline, diesel, jet fuel, distillate, distillate fuel, residual fuel
- refined products, refining margin, refinery throughput, refinery utilization
- Strategic Petroleum Reserve, SPR, SPR release
- WTI, Brent, OPEC oil production, OPEC+ oil quota, OPEC oil cuts
- "barrels per day" or "b/d" or "bpd" when referring to oil/crude/petroleum production or consumption

EXCLUDED TERMS (these alone do NOT count as oil-market triggers):

- natural gas, LNG, liquefied natural gas, Henry Hub, Bcf/d, Mcf, gas storage, gas pipeline (NOT an oil trigger)
- coal, coal-fired generation, coal capacity, coal retirement, coal mining (NOT an oil trigger)
- electricity, electric power, power generation, power sector demand, kilowatthour (NOT an oil trigger)
- renewable, solar, wind, battery storage, hydroelectric, nuclear (NOT an oil trigger)
- biofuels, ethanol, RINs, RFS (NOT an oil trigger unless the passage explicitly links them to gasoline blending or oil substitution)
- propane, butane, natural gas liquids (NGLs) -- these are EXCLUDED unless the passage discusses them as substitutes for oil products specifically

If a passage discusses BOTH oil and a non-oil commodity (e.g., "rising natural gas prices increased demand for oil-based distillate fuel"), include the reason -- but the explanation must reference the oil/petroleum/distillate side of the linkage, not the natural-gas side alone. The "reason" label should describe the oil-side effect.

If the passage only discusses non-oil commodities (e.g., "natural gas prices rose due to LNG exports"), EXCLUDE the reason regardless of how oil-relevant the macro narrative might be. Do not infer oil-market relevance from non-oil text.

If the explicit oil-market reference is not present in the quoted passage itself, exclude the reason -- even if economic reasoning could connect it to oil markets through cross-fuel logic.

Rule 2 (Structural cause): A reason must be a structural driver, not a tautological or self-referential restatement of the outcome. Do NOT include:

- "rising oil prices" without a structural cause
- "tight oil markets" without explanation
- "oil price volatility" as a reason in itself
- "expected price increases" without an underlying driver

DO include structural drivers such as:

- "Chinese economic recovery driving oil demand"
- "OPEC+ production cut announcement"
- "Russian sanctions on oil exports"
- "Atlantic hurricane season impact on Gulf oil production"
- "U.S. shale oil capacity expansion"
- "Long-run substitution from gasoline to electric vehicles"
- "Strategic Petroleum Reserve releases"
- "Refinery maintenance reducing distillate output"

Rule 3 (Multiple reasons per passage): When a single sentence or passage contains multiple distinct structural drivers (e.g., "OPEC cuts and stronger Chinese oil demand are both pushing crude prices higher"), extract and list each as a separate entry. Do not merge.

=====

EDGE CASES TO RESOLVE

=====

(a) Cross-fuel substitution effects on oil:

INCLUDE if the passage explicitly says oil/distillate/gasoline demand changed because of substitution.

Example: "High natural gas prices led utilities to switch to fuel oil for power generation, raising distillate demand" -- INCLUDE because the passage references distillate demand. The reason label should be about the oil side.

EXCLUDE if the passage describes substitution that does not affect oil. Example: "High natural gas prices led utilities to switch to coal" -- EXCLUDE because oil/petroleum is not affected.

(b) Macroeconomic drivers (GDP, recession, China growth):

INCLUDE only if the passage explicitly links the macro driver to oil/petroleum/refined products. Example:

"U.S. GDP growth of 2.5% supports our oil consumption forecast" -- INCLUDE.

EXCLUDE if the passage discusses GDP only with reference to natural gas, electricity, or coal. Example:

"U.S. GDP growth supports natural gas consumption" -- EXCLUDE.

(c) Long-run substitution toward non-oil energy:

INCLUDE if the passage says EV adoption, efficiency gains, or non-oil substitution will reduce gasoline/oil demand. Example: "Continued EV adoption is expected to reduce U.S. gasoline consumption growth through 2030" -- INCLUDE because gasoline demand is the affected variable.

EXCLUDE if the passage discusses non-oil energy transitions without explicit oil-side effects. Example:

"Renewable capacity additions will displace coal-fired generation" -- EXCLUDE because oil is not in the passage.

(d) Inventory and storage:

INCLUDE petroleum/oil/crude/distillate/gasoline inventory dynamics. EXCLUDE natural gas storage, working gas in storage, etc.

(e) Refining sector:

INCLUDE refining-side dynamics that affect crude demand or refined product output. The refinery channel IS an oil-market channel. Example: "Refinery maintenance season will reduce crude inputs by 1 mb/d" --

INCLUDE.

=====

FIELDS TO RECORD FOR EACH REASON

=====

For each identified reason, populate:

1. "reason": A brief structural-driver label specifically describing the oil-side mechanism (e.g., "Chinese oil demand growth", "OPEC quota cut", "Hurricane oil disruption"). Make it clear from the label that the driver is oil-related.
2. "explanation": The full sentence or passage from the report that supports the reason. Wrap the quoted portion in single quotes ('...') so JSON remains valid. The quoted passage MUST itself contain at least one ALLOWED OIL-MARKET TERM from Rule 1. Do not paraphrase a non-oil passage to make it sound oil-relevant -- you must quote the actual oil-market language from the source.
3. "importance_score": An integer from 0 to 10 reflecting how heavily the report emphasizes this driver at this time.
4. "effect_price": The expected directional effect on oil prices (crude or refined products).
 - "I" = increasing
 - "D" = decreasing
 - "na" = unclear or unspecified
5. "effect_quantity": The expected directional effect on physical oil quantities (production, consumption, or inventories).
 - "I" = increasing
 - "D" = decreasing
 - "na" = unclear or unspecified
6. "horizon": The temporal horizon over which the effect operates.
 - "short" = effects within roughly 12 months
 - "long" = effects unfolding over more than 12 months, including capacity adjustments, behavioral changes, fuel substitution that takes years to play out
 - "na" = unclear
7. "is_forecast": Whether the reason is presented as a current/observed condition or as a forward-looking forecast.
 - "current" = describes a condition the EIA observes now
 - "forecast" = describes a condition the EIA expects to occur
 - "both" = the passage discusses both
 - "na" = unclear

=====

TOKEN AND COMPLETENESS RULES

=====

- If the report contains no reasons that satisfy Hard Inclusion Rules 1-2, return an empty "reasons" array and set "all_reasons" = "Y".
 - If you have identified all relevant oil-specific reasons, set "all_reasons" = "Y".
 - If token or space constraints prevent listing every relevant reason, set "all_reasons" = "N" and list as many as possible in order of importance_score (highest first).
- =====

OUTPUT FORMAT

=====

CRITICAL: Your entire response must be a single JSON object and nothing else. Do not include any preamble, explanation, code fences, or trailing text. The first character of your response must be '{' and the last character must be '}'.

Use exactly this structure:

```
{
  "reasons": [
    {
      "reason": "<brieft oil-side label>",
      "explanation": "<full passage with quoted text in single quotes; quoted passage must contain an
allowed oil-market term>",
      "importance_score": <integer 0-10>,
      "effect_price": "<I/D/na>",
      "effect_quantity": "<I/D/na>",
      "horizon": "<short/long/na>",
      "is_forecast": "<current/forecast/both/na>"
    }
  ],
  "all_reasons": "<Y/N>"
}
```

Example of an INCLUDED reason (oil-market trigger present):

```
{"reason": "OPEC+ production cut", "explanation": "The report notes that 'OPEC+ announced an extension of
voluntary crude oil production cuts of 2.2 million barrels per day through Q2 2024, which is expected
to keep crude oil prices elevated relative to our previous forecast'.", "importance_score": 8,
"effect_price": "I", "effect_quantity": "D", "horizon": "short", "is_forecast": "forecast"}
```

Example of an EXCLUDED reason (no oil-market trigger; would have been wrongly included by a broad reading):

The passage "EIA expects U.S. natural gas consumption will average 79.1 Bcf/d in 2021, a 4.3% decline from 2020. The expected decline is the result of rising natural gas prices that will reduce demand for natural gas in the electric power sector." discusses ONLY natural gas. There is no oil/crude/petroleum/distillate/gasoline reference. EXCLUDE.

Example of an INCLUDED cross-fuel reason (oil reference present):

```
{"reason": "Distillate substitution from natural gas in European power", "explanation": "The report states
that 'rapidly increasing natural gas prices in Europe and Asia have increased demand for distillate
fuel for power generation, supporting global distillate prices and consumption'.", "importance_score":
6, "effect_price": "I", "effect_quantity": "I", "horizon": "short", "is_forecast": "forecast"}
```

B.2 Stage 2 system prompt: classification

You are an energy economist classifying a structural reason cited in an EIA Short-Term Energy Outlook report. Use ONLY the provided reason label, explanation passage, and effect fields. Do not draw on external knowledge about specific historical episodes, OPEC meetings, or other events not described in the provided context.

GOAL

Classify the reason into exactly ONE category from the following set (case-sensitive):

1. Aggregate_Demand
2. Oil_Specific_Demand
3. Supply
4. Inventory
5. Substitution_Induced
6. Other

CATEGORY DEFINITIONS AND SIGN RESTRICTIONS

(1) Aggregate_Demand

Macroeconomic activity driving oil consumption through general economic channels. Examples: GDP growth, industrial production, recession, China growth, global economic recovery, manufacturing activity, transportation activity tied to economic conditions.

Sign restrictions:

- Positive shift: price up, quantity (consumption) up
- Negative shift: price down, quantity (consumption) down

(2) Oil_Specific_Demand

Demand specific to oil markets driven by precautionary inventory accumulation, geopolitical risk premia, war risk hedging, sanctions-induced stockpiling, or expected future scarcity. This is demand for the storage value of oil under uncertainty, not consumption demand.

Sign restrictions:

- Positive shift (heightened scarcity fear): price up, quantity (consumption) flat or slightly up via inventory accumulation
- Negative shift (calming fears): price down, quantity flat or slightly down

(3) Supply

Physical oil production drivers including OPEC and OPEC+ actions, country-level production decisions, capacity additions, capacity disruptions, weather-related outages (hurricanes, freezes), sanctions affecting production, technological changes, refinery throughput.

Sign restrictions:

- Positive shift (abundance): price down, quantity (production) up
- Negative shift (tightness/disruption): price up, quantity (production) down

(4) Inventory

Current oil-market inventory dynamics including SPR releases or builds, OECD commercial inventory levels, refinery product stocks, distillate stocks. This category applies when the passage is specifically about realized inventory levels affecting prices, distinct from precautionary demand which is about expected future scarcity.

Sign restrictions:

- Positive draw (inventories falling): price up, quantity (consumption) down or flat
- Positive build (inventories rising): price down, quantity (consumption) up or flat

(5) Substitution_Induced

Long-run feedback effects unfolding over more than 12 months. Examples: demand destruction from sustained high prices, capacity expansions in response to expected high prices, fuel switching, EV adoption, efficiency gains, behavioral changes in consumption.

Apply this category whenever the explanation passage explicitly invokes a long-run dynamic, regardless of whether the short-run sign restrictions match a different category.

Sign restrictions for long-run effects:

- Long-run demand destruction: future quantity down, future price down
- Long-run supply expansion: future quantity up, future price down

(6) Other

Financial flows, currency movements, regulatory changes, refining margins, or causes that don't fit the above. Use as a residual category.

CLASSIFICATION PROCEDURE

Step 1: Read the reason label and explanation passage carefully.

Step 2: Identify which category's sign restrictions match the effect_price and effect_quantity fields provided to you.

Step 3: If the passage explicitly invokes long-run dynamics (effects over 12+ months, behavioral changes, capacity adjustments unfolding over years), classify under Substitution_Induced regardless of short-run sign matching.

Step 4: If multiple categories could match, prefer the more specific category. For example, if a reason involves both physical disruption and precautionary inventory accumulation, classify under Supply if the disruption is the primary driver and Oil_Specific_Demand if the precautionary response is.

Step 5: If the explanation is too vague, tautological, or doesn't fit any category, use Other.

FIELDS TO RECORD

1. "category": One label from the six-category set above (case-sensitive).
2. "scratch": A brief causal chain (1-2 sentences, max 200 chars) explaining your classification reasoning. For example: "OPEC+ cuts physical production, raising prices and lowering quantity supplied. Sign pattern matches Supply (negative)."
3. "explicit": Whether the passage directly mentions oil-market terms.
 - "Y" = passage contains explicit oil-market terms (oil, crude, gasoline, etc.)
 - "N" = passage discusses macro factors that we infer affect oil markets through economic logic
 - "na" = unclear
4. "direction_in_category": The direction of the shift within the assigned category, defined so that "+" indicates an expansion or strengthening of that category's underlying force.
 - "+" = aggregate demand expansion, heightened oil-specific demand, supply abundance, inventory build, long-run supply expansion
 - "-" = aggregate demand contraction, calming of oil-specific demand, supply tightness/disruption, inventory draw, long-run demand destruction

OUTPUT FORMAT

Return strictly valid JSON. Do NOT wrap in markdown code fences. Do NOT include any text before or after the JSON.

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{"category": "<one of the six labels>", "scratch": "<1-2 sentence causal chain>", "explicit": "<Y/N/na>", "direction_in_category": "<+/->"}
```

B.3 Construction of the imbalance series

From the reason-level output, the baseline series are built as follows. Reasons with a Stage 2 error, an unparseable report date, or a direction other than + or − are dropped. Reasons are retained if their importance score is at least five and their explicit-reference flag is Y, so that every retained reason’s quoted passage names an oil-market variable directly. Within each report, $N_{c,t}^+$ and $N_{c,t}^-$ count retained reasons in category c with direction + and −, and the imbalance is $I_{c,t} = (N_{c,t}^+ - N_{c,t}^-) / \sum_{c'} (N_{c',t}^+ + N_{c',t}^-)$, where the sum in the denominator runs over all six categories including the residual. Quarterly reports before 1998 are assigned to the first month of the quarter and carried forward. Three alternative normalizations are computed for robustness: an importance-weighted version in which each reason is weighted by its score divided by ten, in both numerator and denominator; a moving-average version in which the denominator is replaced by the mean number of retained reasons over the preceding eight reports; and an inverse hyperbolic sine of the net count with no denominator. Horizon-conditional versions restrict the counts to reasons tagged short or long. The series are then seasonally adjusted as described in Section 5.4.

B.4 Time-windowed newspaper dictionaries

For the attention composites of Section 4.1 we also constructed a version with no look-ahead, in which a third model call converted each report’s retained aggregate-demand and supply reasons, split by direction, into eight to fifteen journalist-style search phrases, and each month’s phrases were applied only to articles published between that report and the next. The phrases were required to be substantive drivers rather than generic oil vocabulary, two to four words long, directional, and to include named entities where the reasons named them. The resulting monthly article counts were too sparse to produce usable series, and the composites in the paper use the fixed dictionaries of Table A2. The prompt is available on request.